NAVTTC × NETSOL Technologies

NIAI — National Initiative for Artificial Intelligence

Complete Kali Linux Installation Lab Guide

VirtualBox on Windows — From Zero to Hacking-Ready

Cybersecurity Fundamentals — Lab Module

A Hands-On, Step-by-Step Guide for Absolute Beginners

2026

Table of Contents

Table of Contents 2
1. Introduction 4
1.1 What is Kali Linux? 4
1.2 What is Oracle VirtualBox? 4
1.3 What is an ISO File? 5
1.4 System Requirements 5
2. Downloading Kali Linux 6
2.1 Finding the Official Website 6
2.2 Choosing Your Download Method 6
2.3 Downloading the ISO 7
2.4 Pre-Built VM Images (Alternative Method) 8
2.5 Download in Progress 9
3. Downloading & Installing VirtualBox 11
3.1 Finding VirtualBox 11
3.2 Downloading VirtualBox 11
3.3 Downloading the Installer 12
3.4 Verifying Downloads 12
3.5 Installing VirtualBox 13
4. Creating the Virtual Machine 14
4.1 Opening VirtualBox Manager 14
4.2 Naming the VM 14
4.3 Attaching the ISO File 15
4.4 Hardware Configuration (RAM & CPU) 17
4.5 Virtual Hard Disk 17
4.6 Review & Create 18
4.7 VM Created 19
5. Booting the VM & Starting Installation 20
5.1 Starting the VM 20
5.2 Kali Boot Menu 21
6. Kali Linux Installation Process 23
6.1 Select Language 23
6.2 Select Location 23
6.3 Configure Keyboard 24
6.4 Network — Hostname 25
6.5 Network — Domain Name 26

6.6 User Account Setup 27
6.7 Configure the Clock 30
6.8 Disk Partitioning 32
6.9 Proxy Configuration 37
6.10 Software Selection 37
6.11 GRUB Boot Loader 39
6.12 Installation Complete! 41
7. Post-Installation Setup 42
7.1 Update the System 42
7.2 Verify Kali Version 42
7.3 Install Additional Tools (Optional) 42
7.4 Take a Snapshot 42
8. Common Issues & Troubleshooting 44
8.1 VT-x/AMD-V Not Available 44
8.2 VM is Very Slow 44
8.3 Black Screen After Boot 44
8.4 No Internet Inside Kali 44
8.5 VERR_PATH_NOT_FOUND Error 44
8.6 ISO Opens with WinRAR 44
8.7 Shared Clipboard Not Working 44
9. Quick Reference Card 45
10. Conclusion 46

1. Introduction

Welcome to the Kali Linux Installation Lab Guide, developed as part of the NIAI (National Initiative for Artificial Intelligence) program — a collaborative effort between NAVTTC (National Vocational & Technical Training Commission) and NETSOL Technologies. This guide is designed to provide participants with practical, hands-on experience in setting up a professional penetration testing environment.

By the end of this guide, you will have a fully functional Kali Linux virtual machine running on your Windows computer, ready for cybersecurity labs and security assessments.

This guide assumes absolutely no prior experience with Linux, virtual machines, or command-line tools. Every single click is documented with a screenshot and a clear explanation of what is happening and why.

1.1 What is Kali Linux?

Kali Linux is a Debian-based Linux distribution specifically designed for digital forensics, penetration testing, and security auditing. It is maintained by OffSec (formerly Offensive Security) and comes pre-loaded with over 600 security tools including Nmap, Wireshark, Metasploit, Burp Suite, John the Ripper, and many more.

> **Fun Fact!**
> Kali Linux was originally released in March 2013 as the successor to BackTrack Linux. The name 'Kali' comes from the Hindu goddess of destruction — fitting for a penetration testing OS! It is completely free and open-source, used by security professionals, ethical hackers, and students worldwide. As of 2026, Kali has been downloaded over 100 million times.

1.2 What is Oracle VirtualBox?

Oracle VM VirtualBox is a free, open-source virtualization software that allows you to run an entire operating system inside a window on your current computer. Think of it as a 'computer within a computer.' Your main Windows installation is called the 'host' and Kali Linux running inside VirtualBox is called the 'guest.'

> **Why Use a Virtual Machine?**
> Running Kali in a VM keeps it completely isolated from your main operating system. You can experiment freely, break things, and simply delete the VM and start over. It is the safest and most recommended way to learn penetration testing without risking your actual system. Professional pentesters use this exact setup.

1.3 What is an ISO File?

An ISO file is a complete disc image — a single file containing the entire contents of a CD/DVD. When you download the Kali Linux ISO, you are downloading the full installation disc in digital form. VirtualBox can mount this ISO file as if it were a physical disc inserted into a virtual DVD drive, allowing you to install Kali from it.

> **Fun Fact!**
> The name 'ISO' comes from ISO 9660, the standard file system used for CD-ROM media. Even though physical discs are largely obsolete, the .iso format lives on as the universal way to distribute operating system installers. A typical Kali ISO is around 4.4 GB — that is over 600 tools packed into one file!

1.4 System Requirements

Before starting, make sure your computer meets the following requirements:

Component	Minimum	Recommended
Processor	64-bit (Intel/AMD)	Intel Core i5 or higher
RAM	2 GB	4–8 GB
Storage	20 GB free space	80 GB (SSD preferred)
Virtualization	VT-x / AMD-V enabled	Same (check BIOS)
Internet	Required for downloads	Stable broadband
Host OS	Windows 10/11 (64-bit)	Windows 10/11 (64-bit)

> **Important: Enable Virtualization in BIOS**
> Most modern CPUs support hardware virtualization (Intel VT-x or AMD-V), but it may be disabled by default in your BIOS/UEFI settings. If VirtualBox gives you errors about VT-x/AMD-V being disabled, restart your computer, enter BIOS (usually by pressing F2, F10, DEL, or ESC during boot), and enable virtualization.

2. Downloading Kali Linux

2.1 Finding the Official Website

Step 1: Open your web browser and search for "Kali Linux" on Google. The official website is kali.org — always make sure you are downloading from the official source.

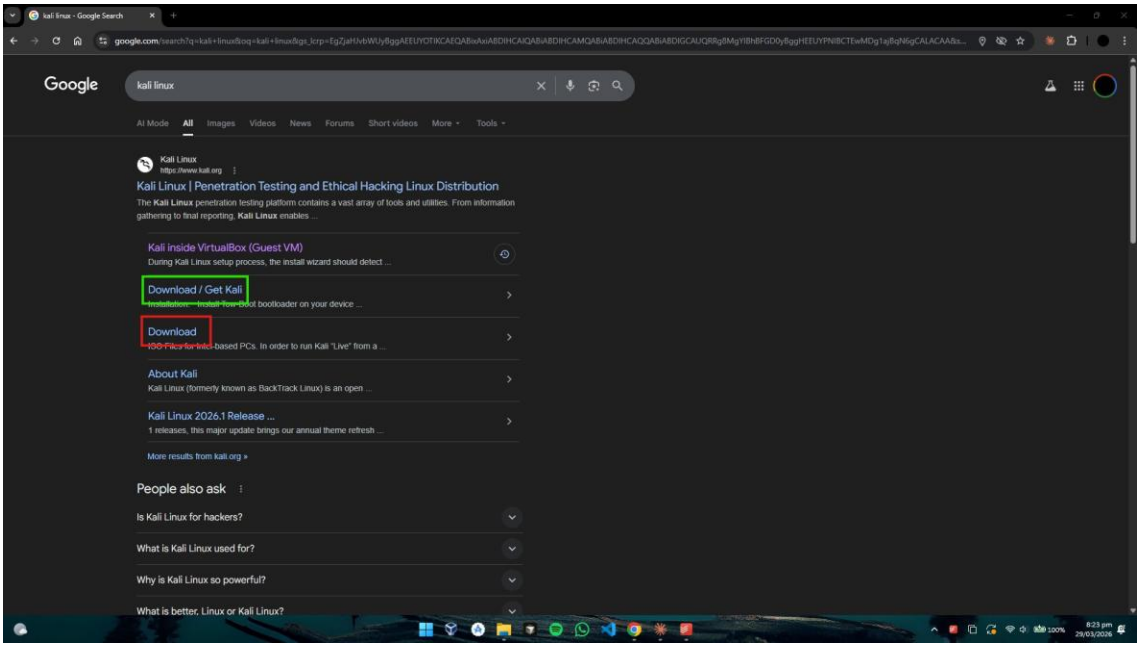

Figure 2.1 — Google search for "Kali Linux." The highlighted links lead to the official download page.

> **Security Warning**
> Never download Kali Linux from unofficial sources, random torrents, or third-party mirrors. Tampered images could contain malware or backdoors. Always use kali.org as your source.

2.2 Choosing Your Download Method

Step 2: Navigate to kali.org/get-kali. You will see the "Choose Your Platform" page. Kali offers multiple ways to get the OS. For this lab, we will use the Installer Images option for a fresh ISO install.

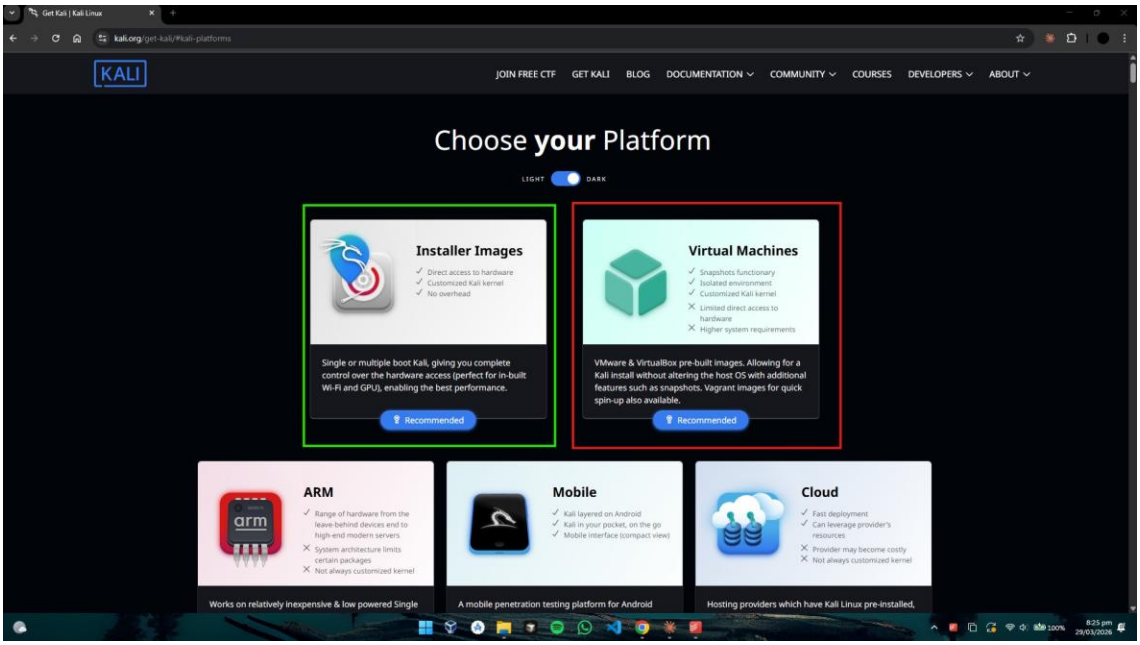

Figure 2.2 — Platform selection page. Green box = Installer Images (ISO). Red box = Virtual Machines (pre-built .ova).

> **Two Ways to Get Kali**
>
> Installer Images (ISO): Download a .iso file and install Kali step-by-step. Gives you full control and teaches the installation process. | Pre-built VMs (.ova): Ready-made VirtualBox images you just import. Faster but less educational. This guide covers the ISO method to give you the complete hands-on experience.

2.3 Downloading the ISO

Step 3: Click "Installer Images." Make sure x86_64 is selected (for standard Intel/AMD processors). Click the download arrow next to "Installer" to begin.

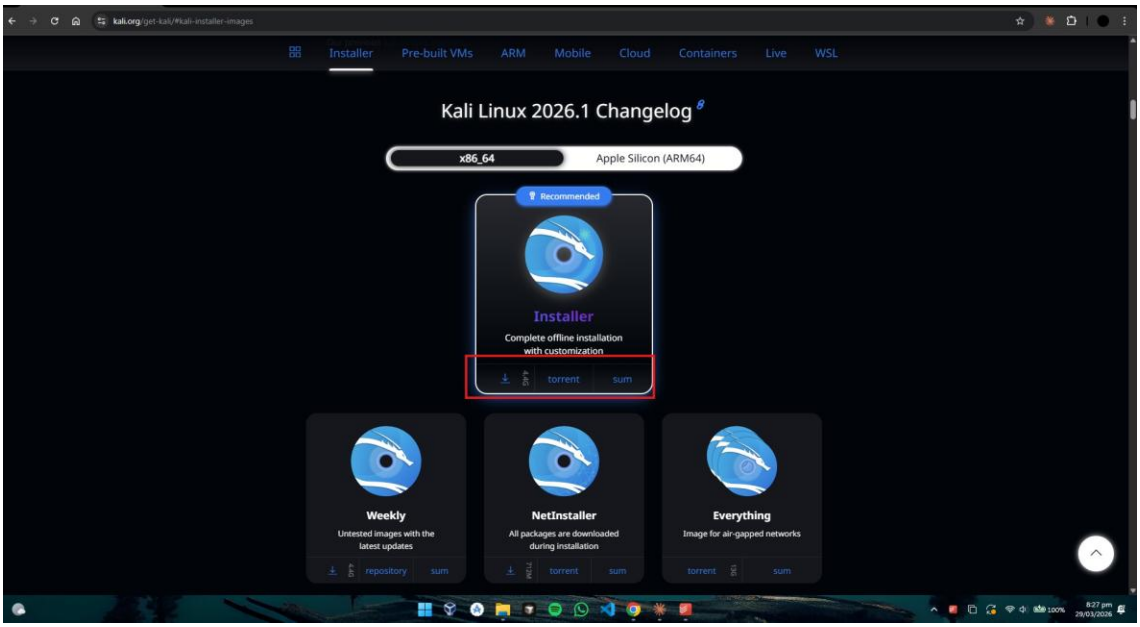

Figure 2.3 — Installer Images page. The red box highlights the download button. File size: approximately 4.4 GB.

> **Fun Fact!**
> The Kali ISO contains over 600 pre-installed penetration testing tools! The "Installer" version includes everything for a complete offline install. The "NetInstaller" is smaller but requires internet during installation to download packages.

2.4 Pre-Built VM Images (Alternative Method)

Step 4: Alternatively, clicking "Pre-built VMs" shows ready-made images for VMware, VirtualBox, Hyper-V, and QEMU. These can be imported directly without going through the installer.

Figure 2.4 — Pre-built VM images. Default credentials: kali/kali.

Default Credentials for Pre-built Images

Pre-built Kali VMs come with default login credentials — Username: kali, Password: kali. Change this immediately after first login for security using the command: passwd

2.5 Download in Progress

Step 5: The ISO file (~4.4 GB) will begin downloading. Do not close the browser until complete.

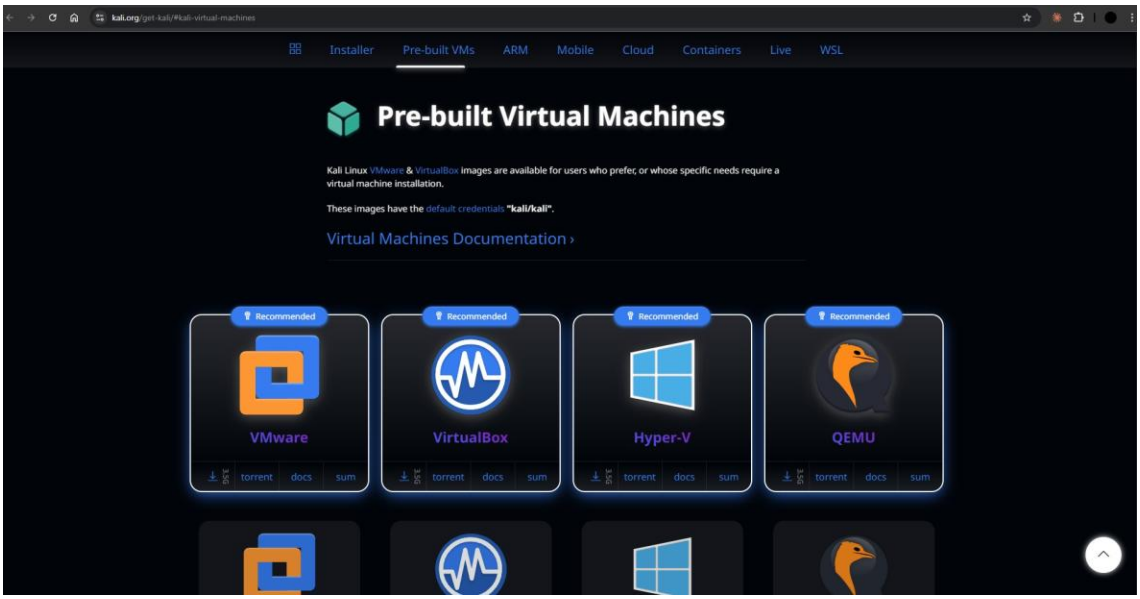

Figure 2.5 — Kali Linux 2026.1 ISO downloading.

3. Downloading & Installing VirtualBox

3.1 Finding VirtualBox

Step 6: Search for "VirtualBox download" on Google and click the official Oracle VirtualBox page.

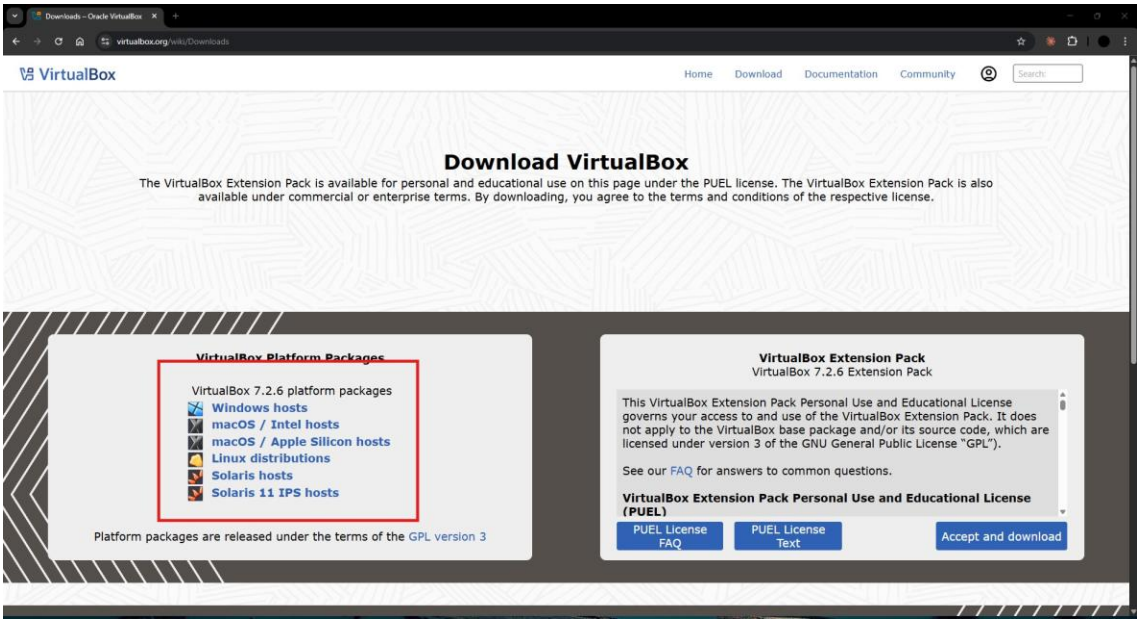

Figure 3.1 — Google search for VirtualBox. Red box highlights the official downloads page.

3.2 Downloading VirtualBox

Step 7: Click "Windows hosts" under VirtualBox Platform Packages to download the installer.

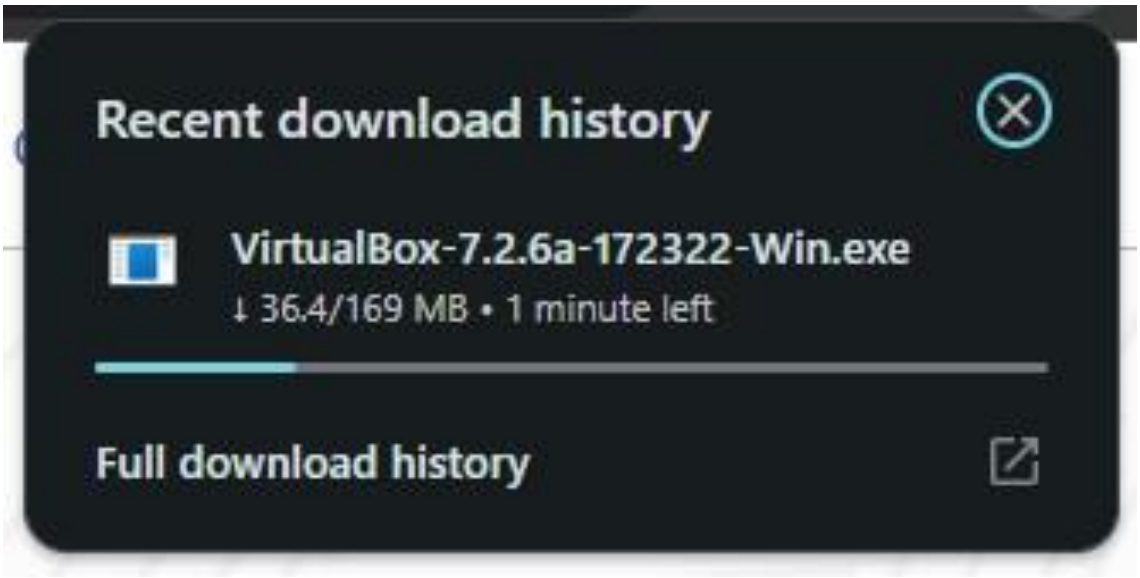

Figure 3.2 — VirtualBox downloads page. Red box highlights the platform packages. Click "Windows hosts."

> **Fun Fact!**
> VirtualBox was created by a German company called Innotek in 2007, acquired by Sun Microsystems in 2008, then Oracle in 2010. Despite being Oracle-owned, it remains free and open-source under GPL. It supports running Windows, Linux, macOS, and even Android as guest operating systems.

3.3 Downloading the Installer

Step 8: The VirtualBox installer (~169 MB) begins downloading.

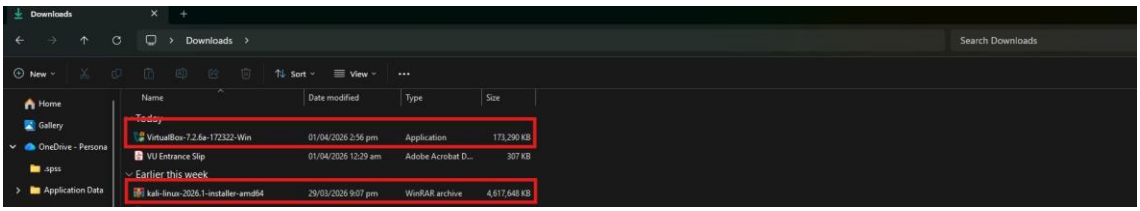

Figure 3.3 — VirtualBox downloading.

3.4 Verifying Downloads

Step 9: Open your Downloads folder. You should see both files: the VirtualBox installer and the Kali ISO.

Figure 3.4 — Downloads folder with VirtualBox installer (173 MB) and Kali ISO (4.6 GB).

3.5 Installing VirtualBox

Step 10: Double-click the VirtualBox installer. Follow the wizard: Next > Next > Install. Accept UAC prompts. Installation takes 1-2 minutes.

> **During Installation**
> You may see warnings about network interfaces being temporarily reset. This is normal — VirtualBox installs virtual network adapters. Your internet will reconnect automatically. If prompted about Python or dependencies, accept defaults.

4. Creating the Virtual Machine

With VirtualBox installed and the Kali ISO downloaded, we now create and configure the virtual machine.

4.1 Opening VirtualBox Manager

Step 11: Open VirtualBox Manager. Click the "New" button (blue star icon) to create a new VM.

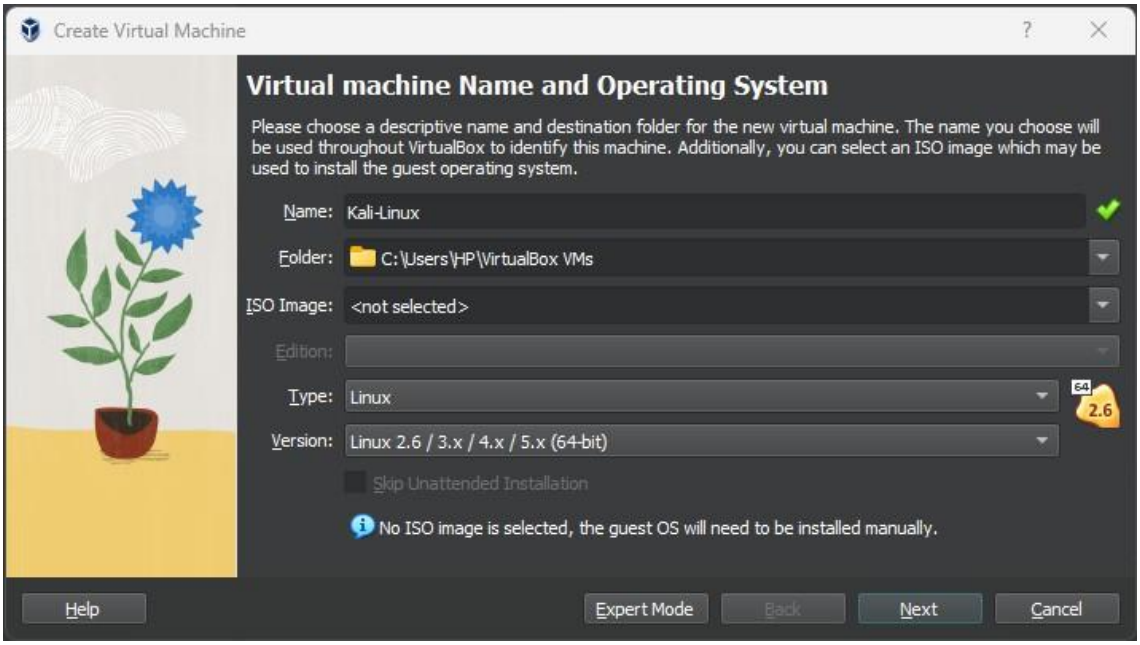

Figure 4.1 — VirtualBox Manager. Red box highlights the "New" button.

4.2 Naming the VM

Step 12: Enter "Kali-Linux" as the Name. Type is set to "Linux" and Version auto-detects once the ISO is attached.

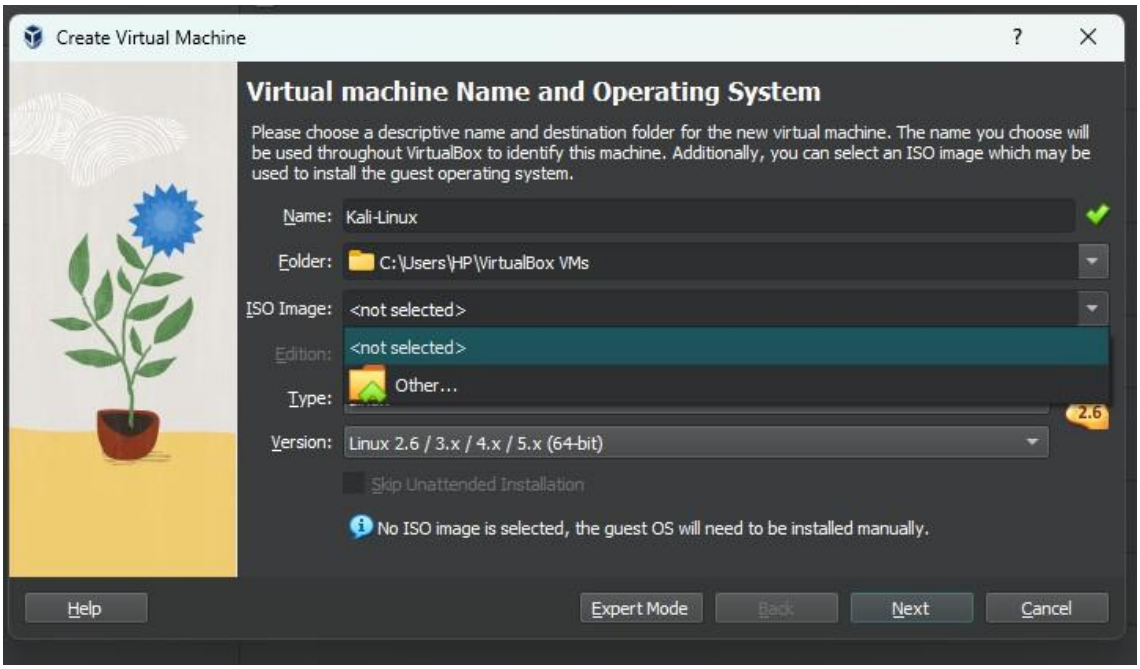

Figure 4.2 — Name and OS screen.

4.3 Attaching the ISO File

Step 13: Click the ISO Image dropdown and select "Other..." to browse for your ISO.

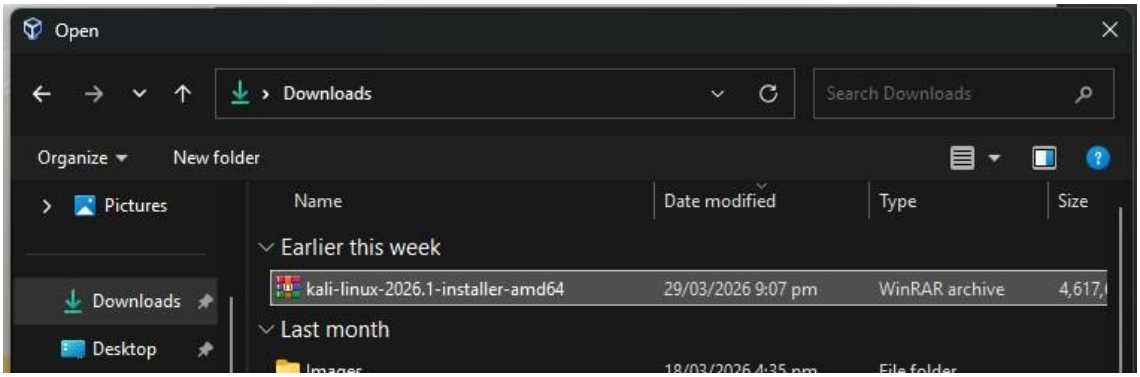

Figure 4.3 — ISO Image dropdown.

Step 14: Navigate to Downloads and select the Kali ISO file.

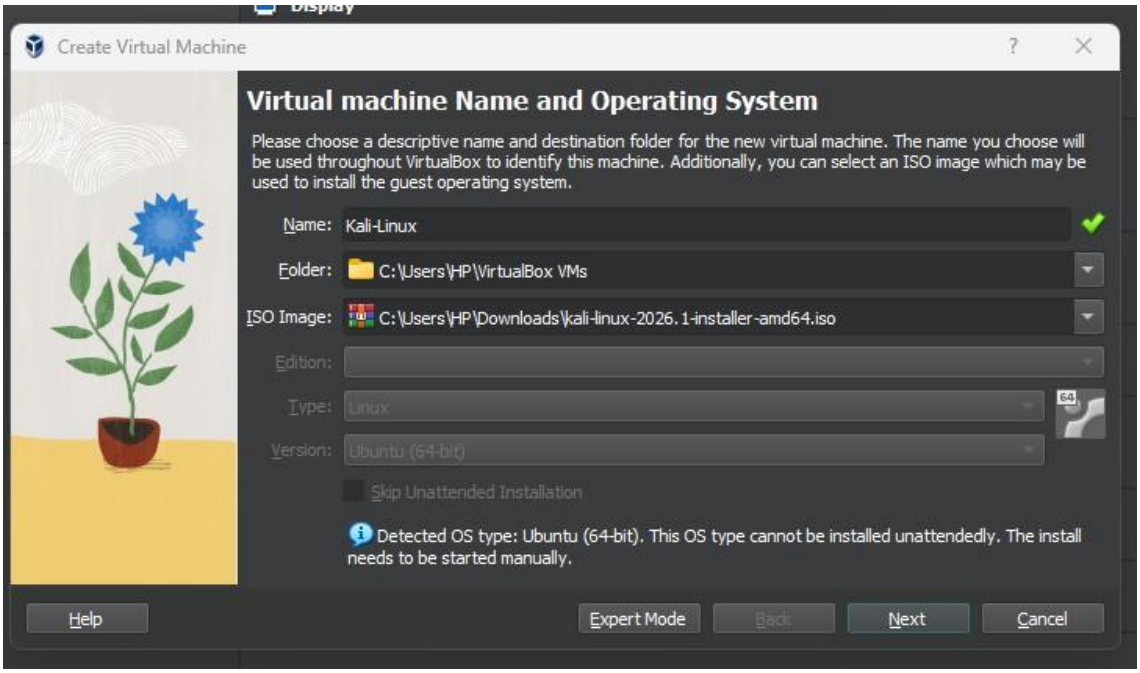

Figure 4.4 — Selecting the Kali ISO from Downloads.

Step 15: VirtualBox auto-detects the OS type as Linux/Ubuntu (64-bit). This is normal for Debian-based distributions like Kali.

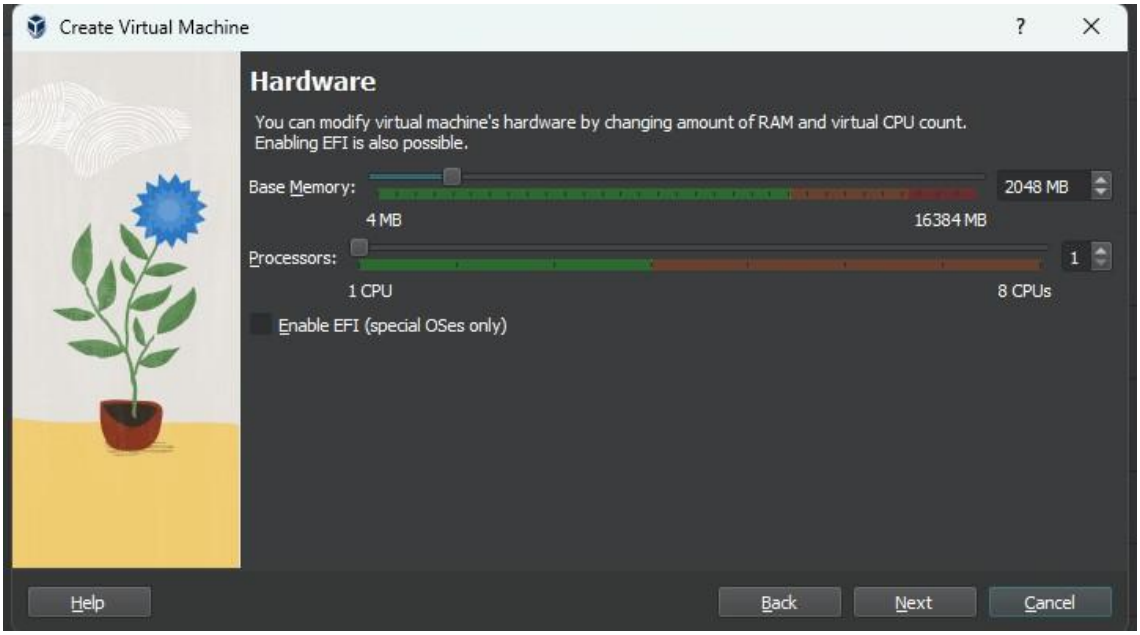

Figure 4.5 — ISO attached. OS detected as Ubuntu 64-bit (correct for Kali).

Why Does It Say Ubuntu?

Kali is based on Debian, and Ubuntu is also Debian-based. VirtualBox auto-detection groups them together. This does not affect functionality at all. The key thing is it detected "Linux" and "64-bit."

4.4 Hardware Configuration (RAM & CPU)

Step 16: Set Base Memory (RAM) to at least 2048 MB. If your system has 8+ GB total, use 4096 MB. Set Processors to 2 if your CPU has 4+ cores.

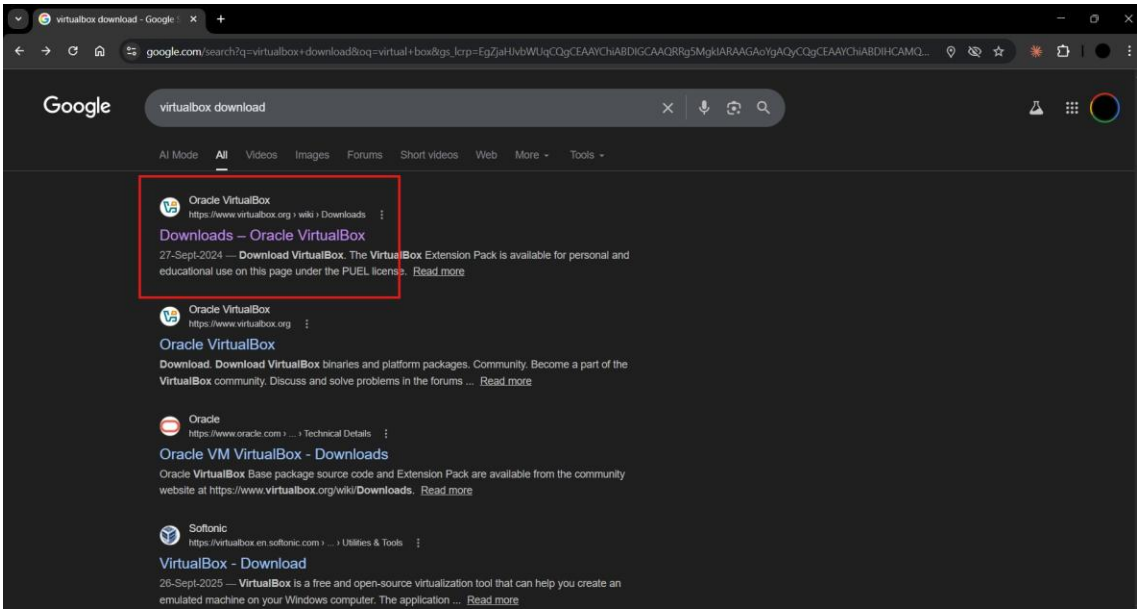

Figure 4.6 — Hardware config: RAM and CPU allocation.

Do Not Over-Allocate!

Never give the VM more than 50% of your total RAM or CPU cores. Over-allocation will make both your Windows host and the VM extremely slow or cause crashes.

4.5 Virtual Hard Disk

Step 17: Select "Create a Virtual Hard Disk Now." Set size to at least 25 GB (50-80 GB recommended). Leave "Pre-allocate Full Size" unchecked for dynamic allocation.

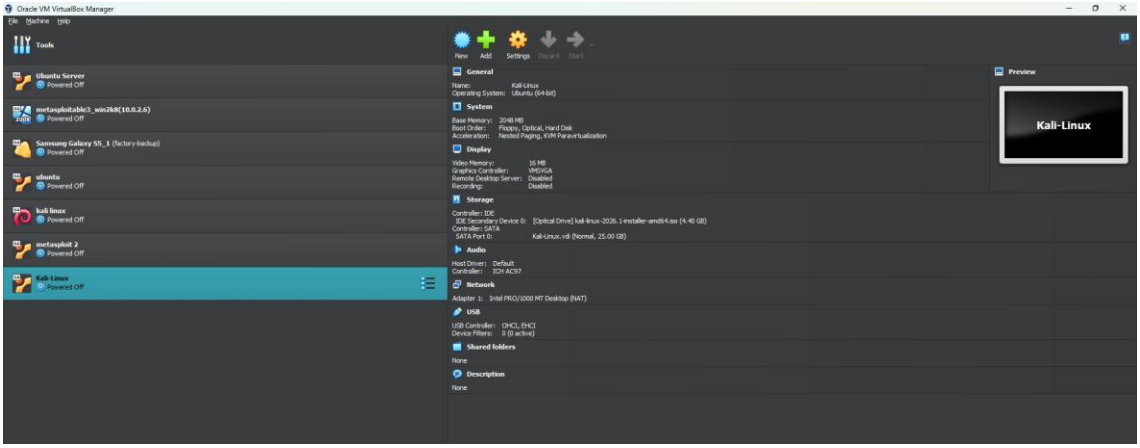

Figure 4.7 — Virtual Hard Disk: 25 GB, dynamically allocated.

> **What is Dynamic Allocation?**
> A dynamically allocated disk starts small and grows only as data is written. An 80 GB dynamic disk might only use 8-10 GB initially. It saves real disk space while giving Kali room to grow.

4.6 Review & Create

Step 18: Review the summary and click "Finish" to create the VM.

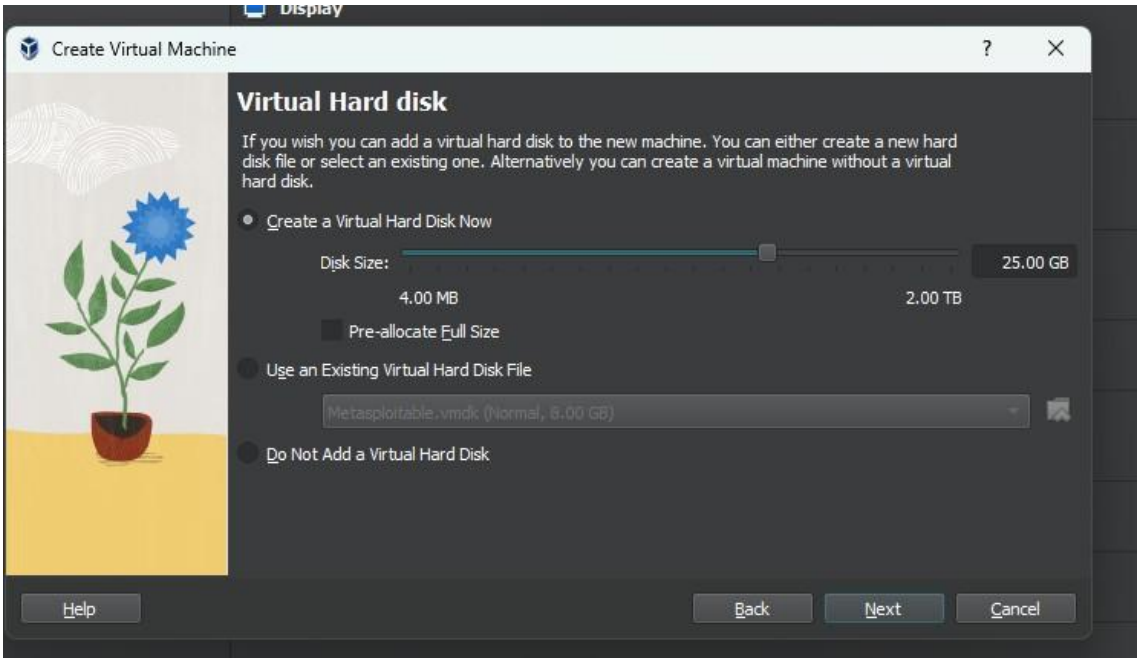

Figure 4.8 — VM configuration summary.

4.7 VM Created

Step 19: Your Kali-Linux VM now appears in the VirtualBox sidebar with all configuration details visible.

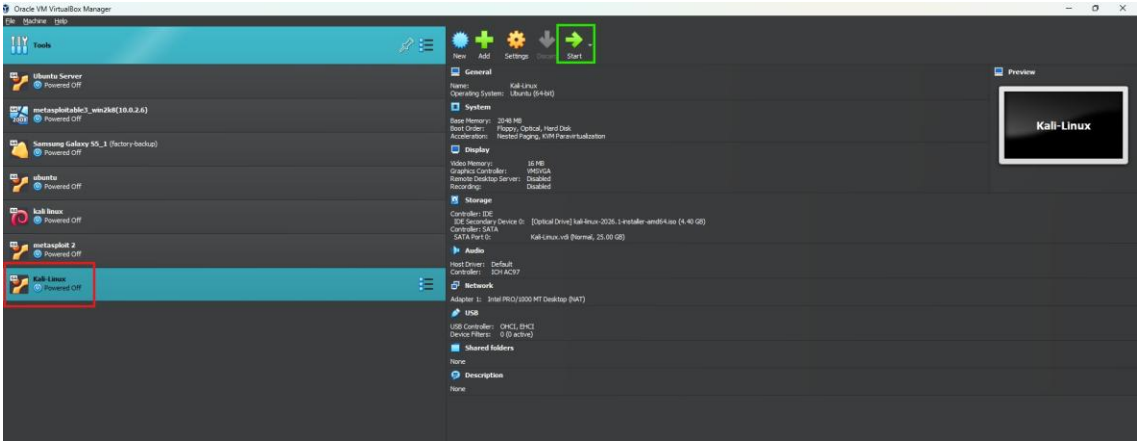

Figure 4.9 — VM created successfully. ISO attached under Storage.

5. Booting the VM & Starting Installation

5.1 Starting the VM

Step 20: Select your Kali-Linux VM and click "Start" (green arrow).

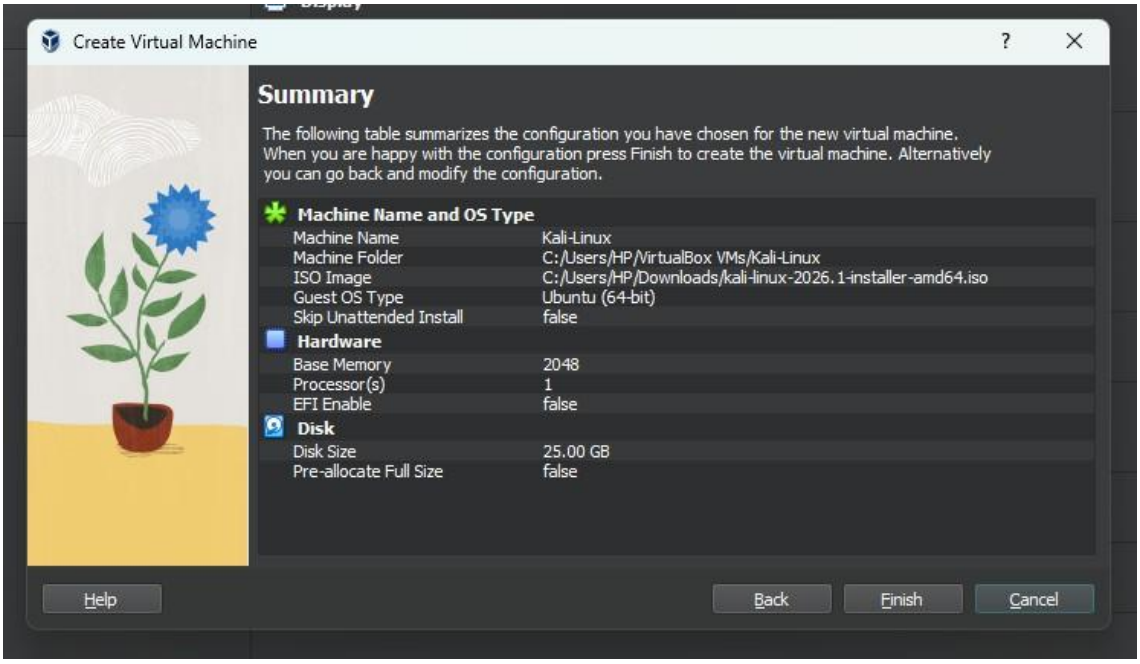

Figure 5.1 — Green box = Start button. Red box = selected VM.

Step 21: VirtualBox powers up the VM.

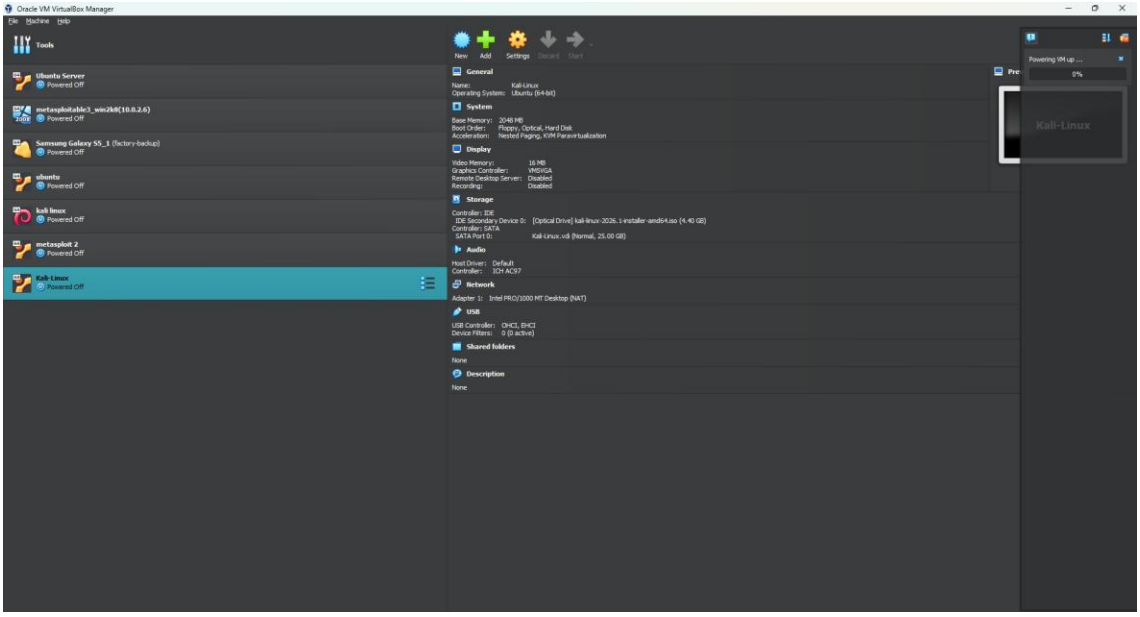

Figure 5.2 — VM powering up.

5.2 Kali Boot Menu

Step 22: Select "Graphical install" (highlighted by default) and press Enter.

Figure 5.3 — Kali Linux boot menu. Select "Graphical install."

Graphical vs Text Install

"Graphical install" gives you a mouse-driven GUI — much easier for beginners. "Install" (text mode) is keyboard-only but produces identical results. Always choose Graphical install.

Mouse Capture

When you click inside the VM window, VirtualBox captures your mouse. Press the Right Ctrl key to release it back to Windows. This "Host Key" is shown in the bottom-right corner of the VM window.

6. Kali Linux Installation Process

The graphical installer walks you through a series of configuration screens. The entire process takes 15-30 minutes. Follow each step carefully.

6.1 Select Language

Step 23: Select "English" (recommended for consistency with security tool documentation) and click Continue.

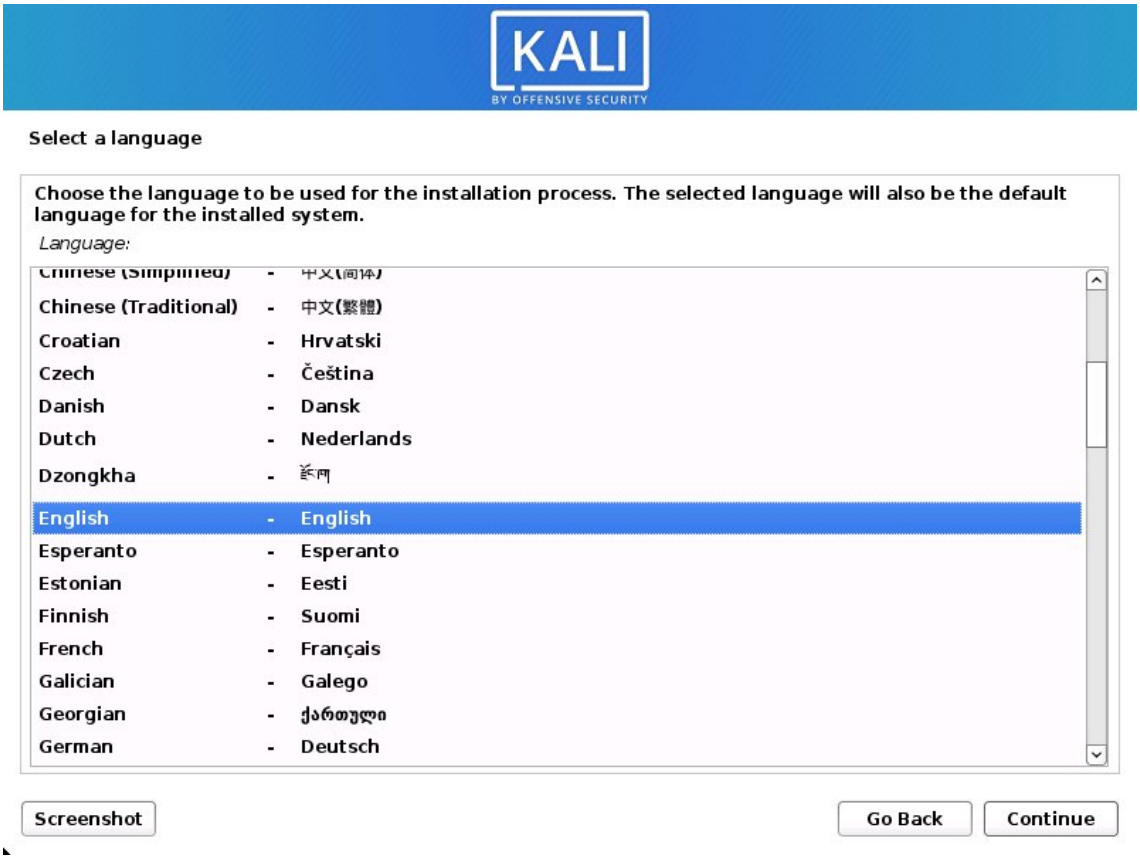

Figure 6.1 — Language selection.

6.2 Select Location

Step 24: Select your country. If not in the shortlist, select "other" at the bottom to browse all regions.

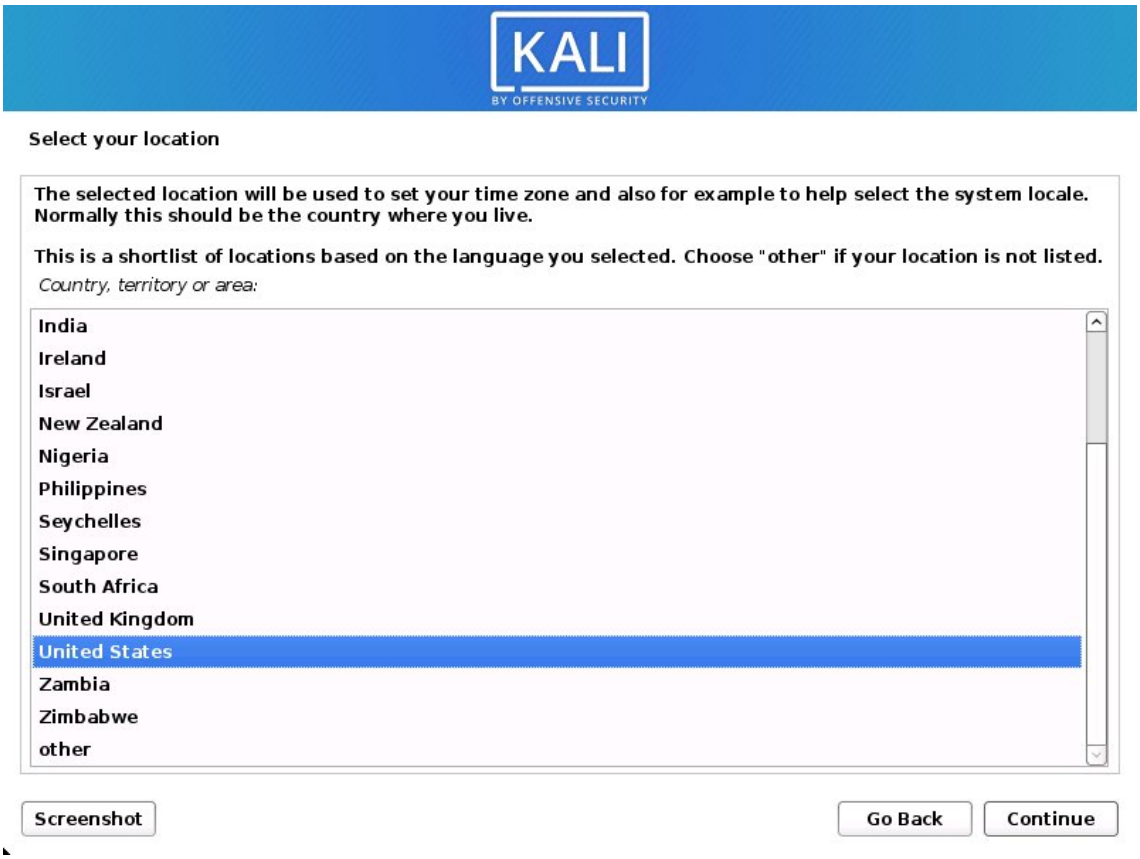

Figure 6.2 — Location selection. Determines your timezone.

6.3 Configure Keyboard

Step 25: Select "American English" (standard QWERTY) and click Continue.

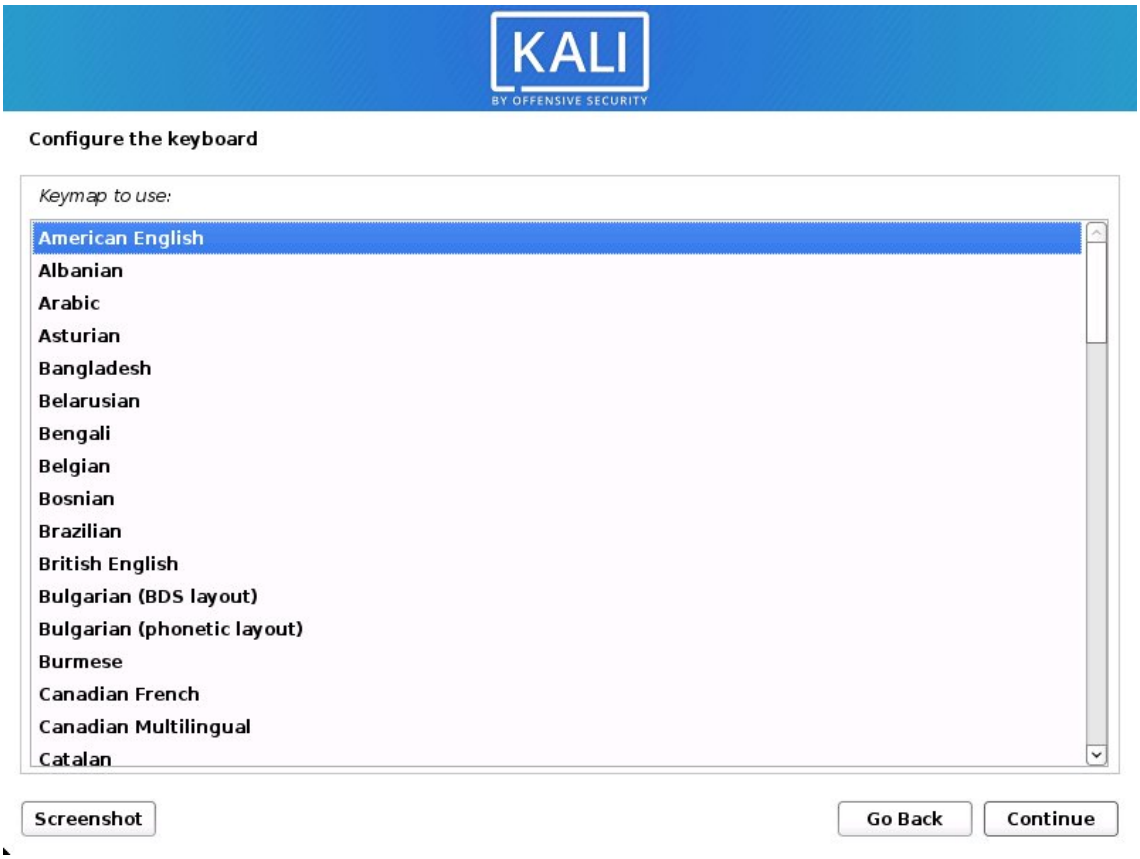

Figure 6.3 — Keyboard configuration.

6.4 Network — Hostname

Step 26: Enter a hostname. The default "kali" is fine for a lab VM. This is how your machine identifies itself on a network.

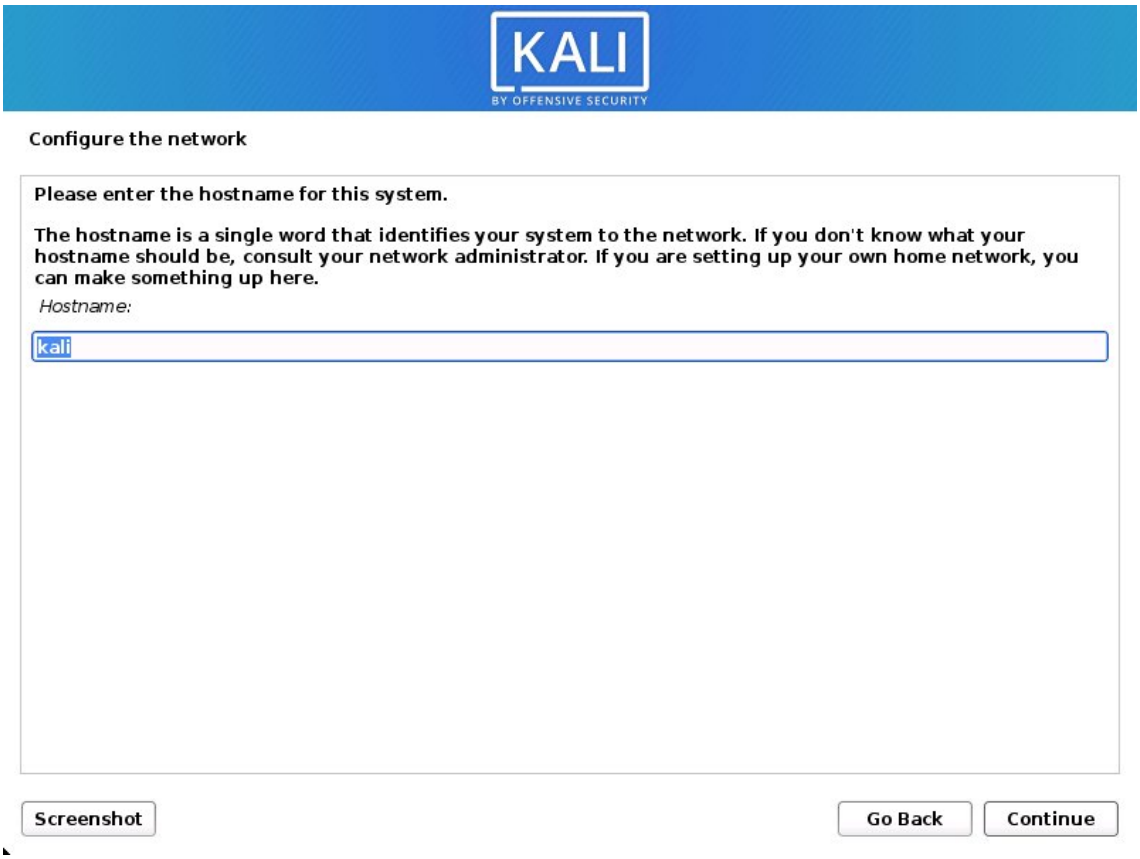

Figure 6.4 — Hostname: "kali"

> **Fun Fact!**
> A hostname is your computer's network nickname. When running tools like Nmap, this name appears in logs. Pentesters often use creative or stealthy hostnames during engagements to blend in with regular network traffic.

6.5 Network — Domain Name

Step 27: Leave the domain name blank and click Continue. Only needed for corporate network environments.

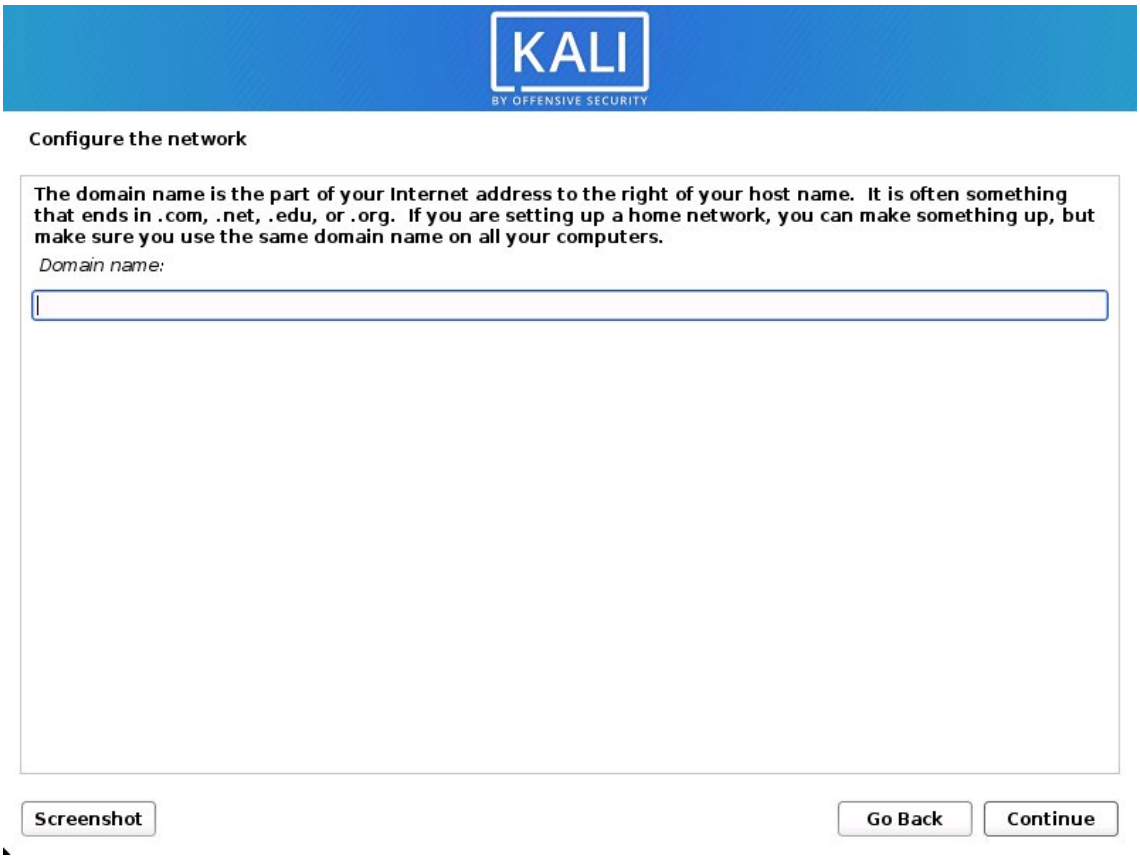

Figure 6.5 — Domain name: leave blank.

6.6 User Account Setup

Step 28: Enter your full name (display name, not login username).

Figure 6.6 — Full name for the user account.

Step 29: Choose a lowercase username for login.

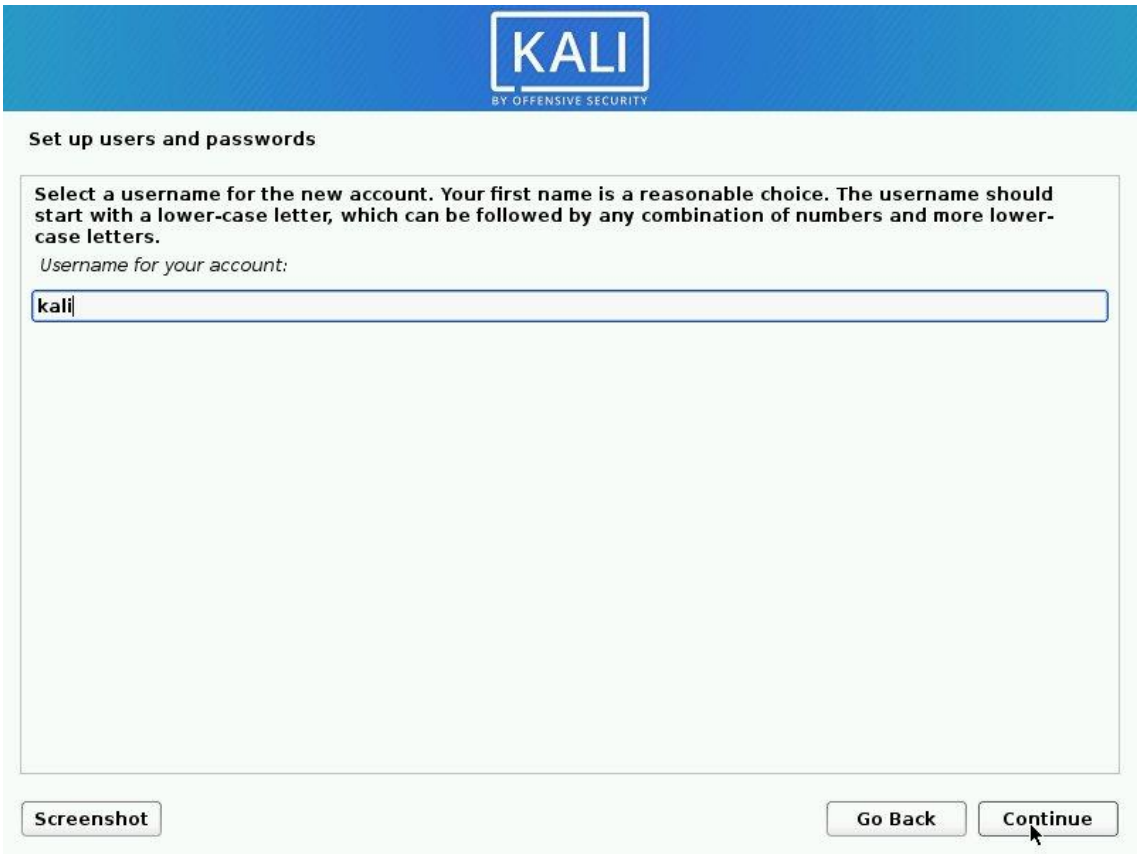

Figure 6.7 — Username selection.

Step 30: Set a strong password. Enter it twice to confirm. This is also used for sudo (admin) commands.

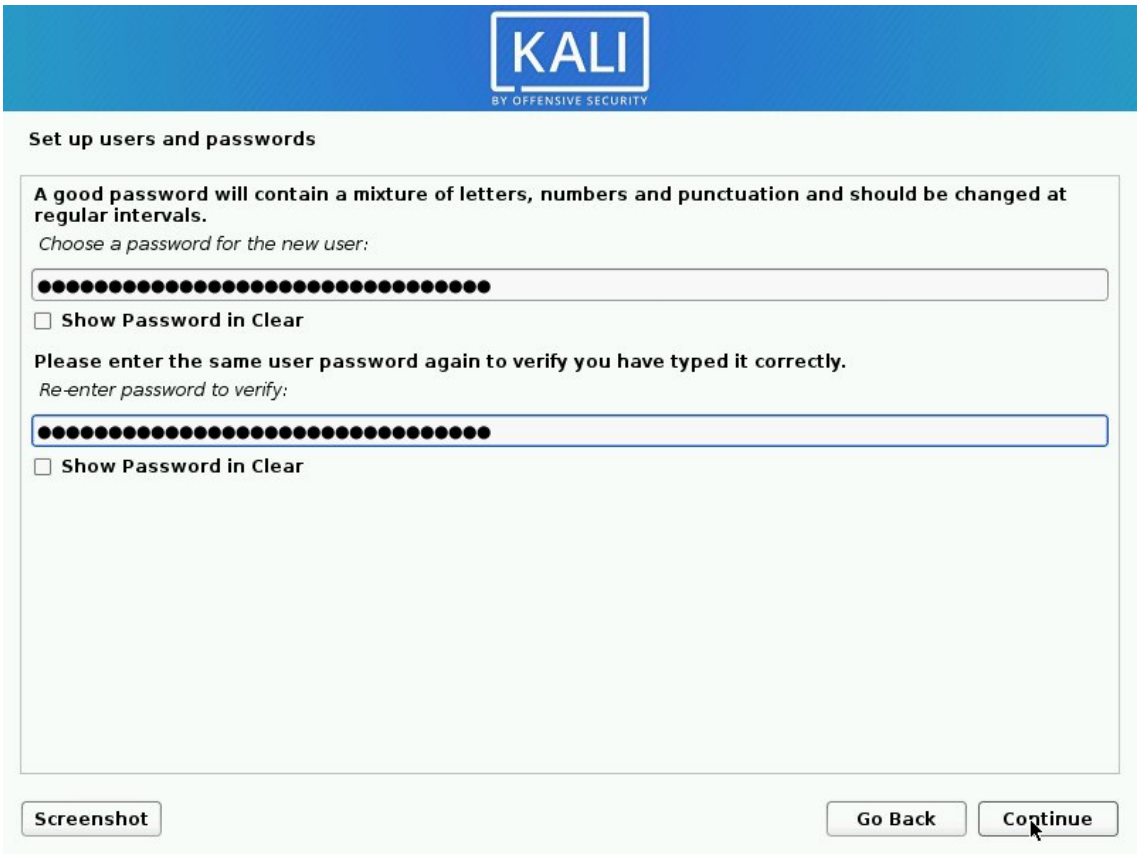

Figure 6.8 — Password setup.

> **Remember Your Password!**
> You will need this password every time you log in and for all sudo commands. If forgotten, system recovery is difficult. Write it down somewhere safe.

6.7 Configure the Clock

Step 31: Select your timezone and click Continue.

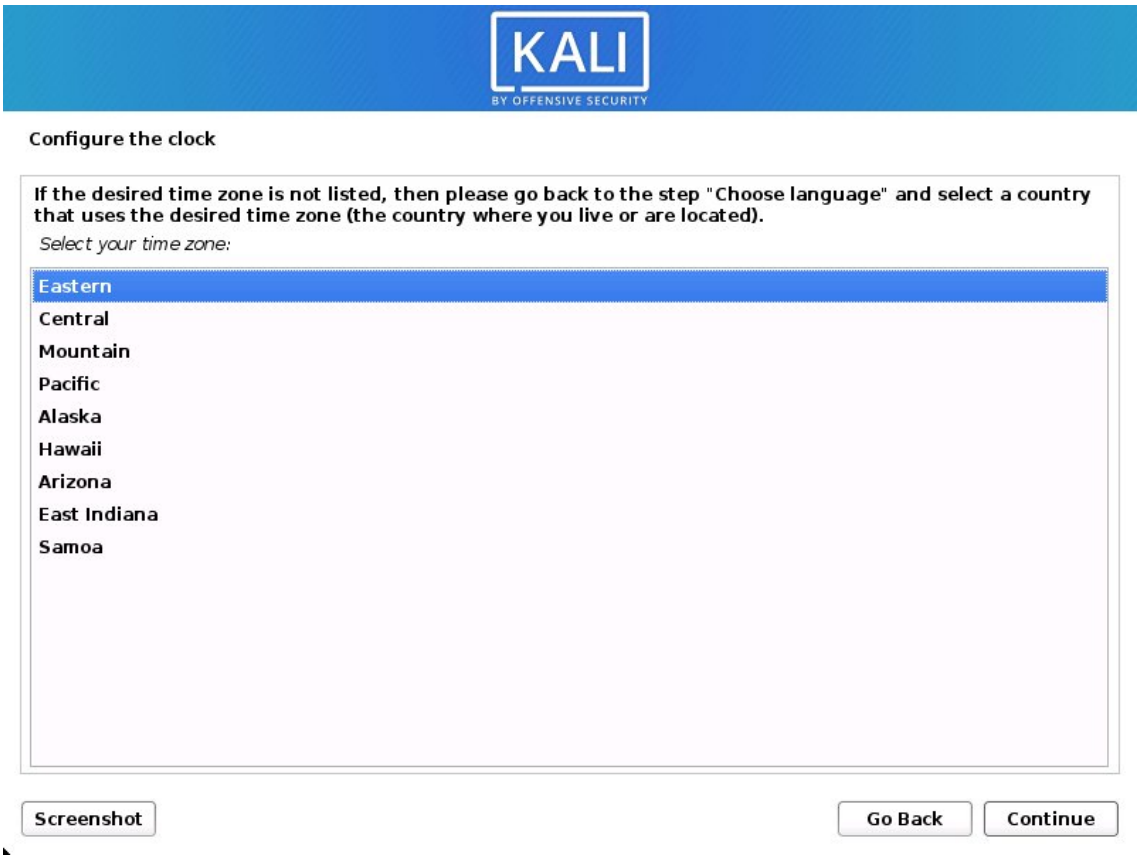

Figure 6.9 — Timezone selection.

6.8 Disk Partitioning

Partitioning sets up the virtual hard disk structure. Since this is a VM, we safely use the entire disk.

Step 32: Select "Guided - use entire disk" and click Continue.

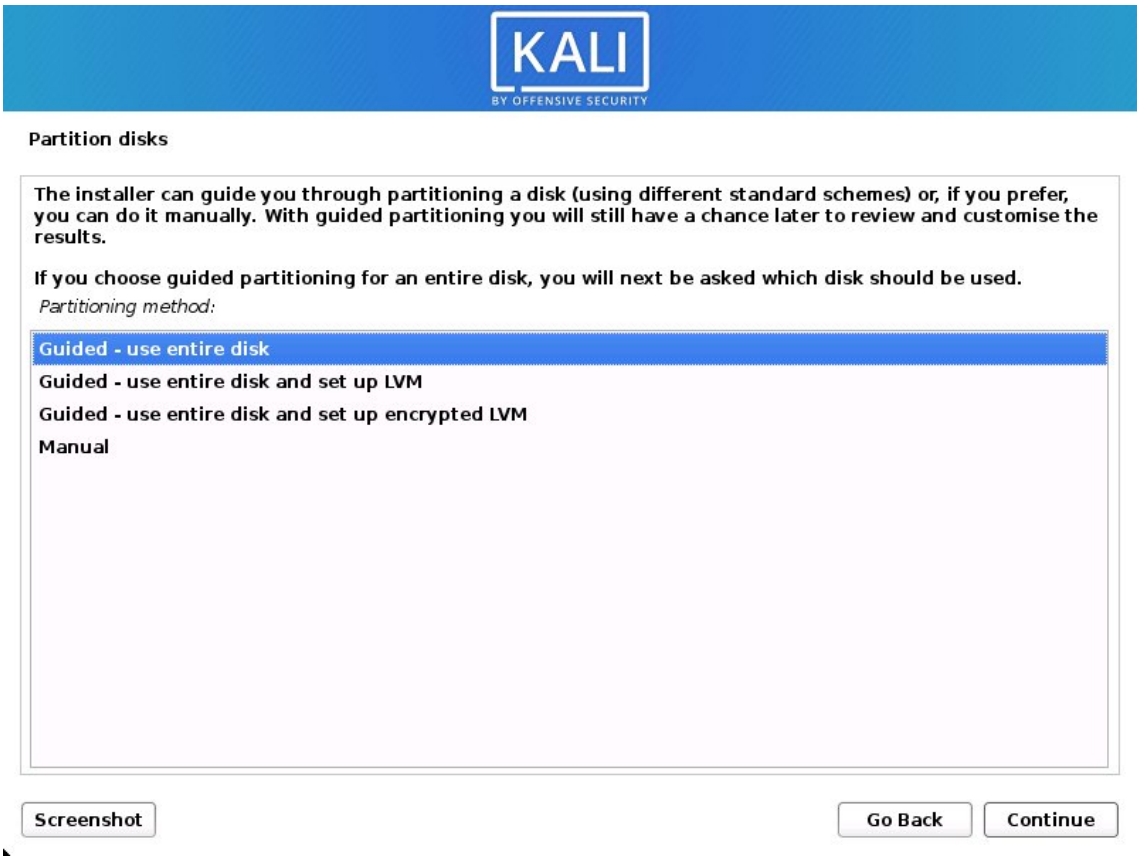

Figure 6.10 — Partitioning method: Guided - use entire disk.

Step 33: Select the only available virtual disk and click Continue.

Figure 6.11 — Disk selection.

Step 34: Select "All files in one partition (recommended for new users)" and click Continue.

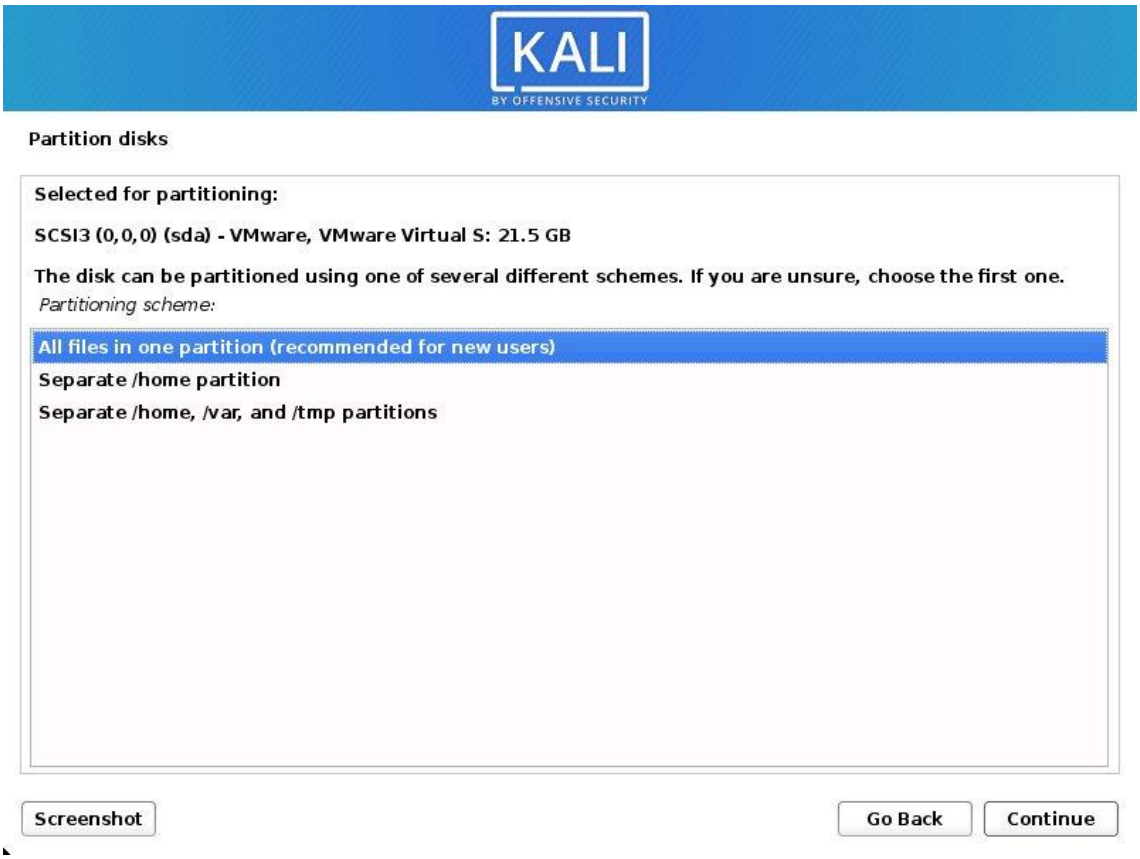

Figure 6.12 — Partitioning scheme: all files in one partition.

Step 35: Review the partition layout. Select "Finish partitioning and write changes to disk" and click Continue.

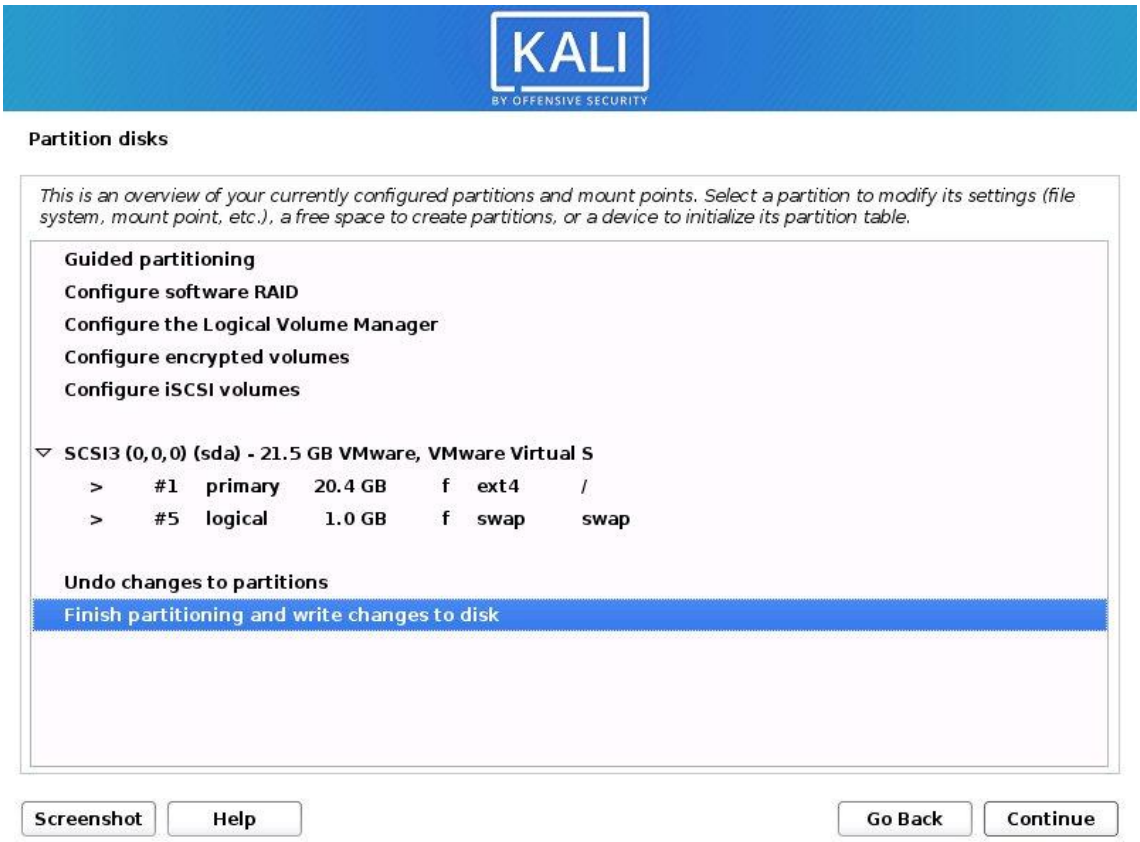

Figure 6.13 — Partition overview: ext4 main partition + swap.

> **Fun Fact!**
> ext4 (fourth extended filesystem) has been the Linux default since 2008. Swap space acts as overflow memory when RAM is full — essentially, your hard disk pretends to be extra RAM, though much slower.

Step 36: Confirm by selecting "Yes" to write changes to disk.

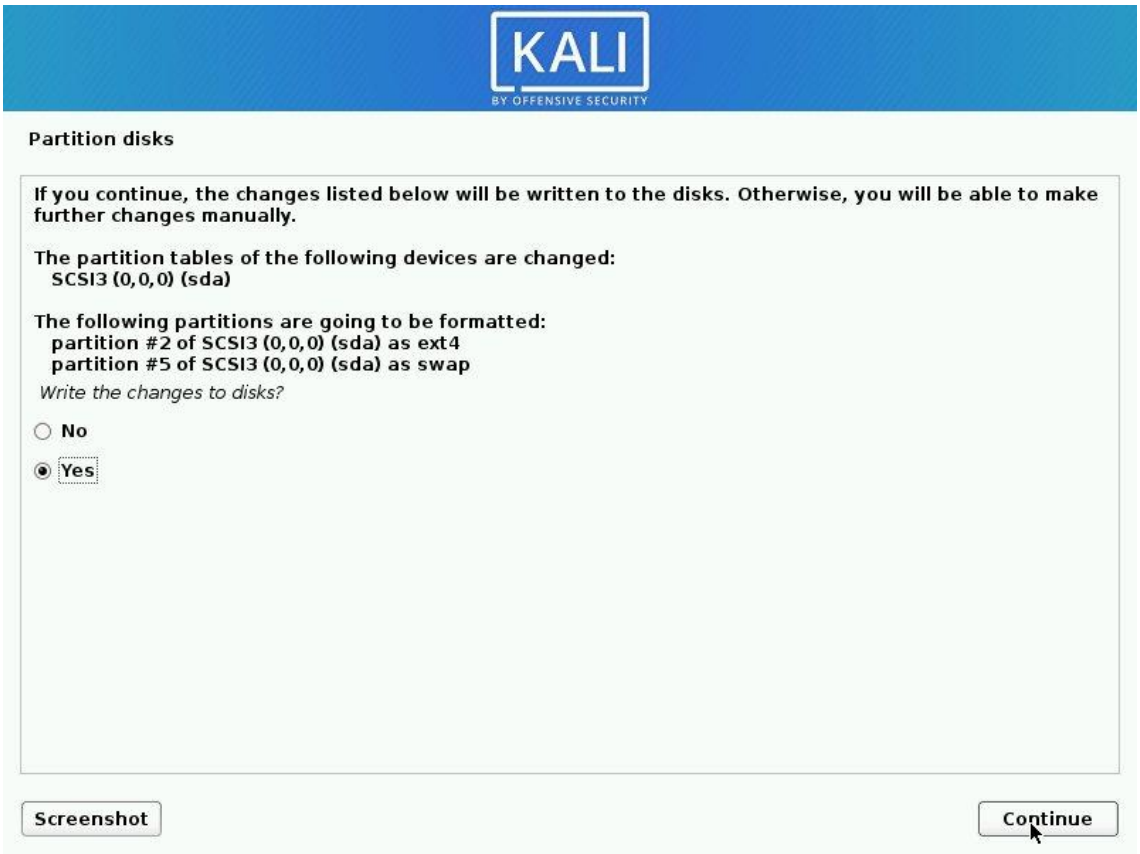

Figure 6.14 — Final confirmation: Yes to write changes.

6.9 Proxy Configuration

Step 37: Leave blank for direct internet access (most home/lab networks). Click Continue.

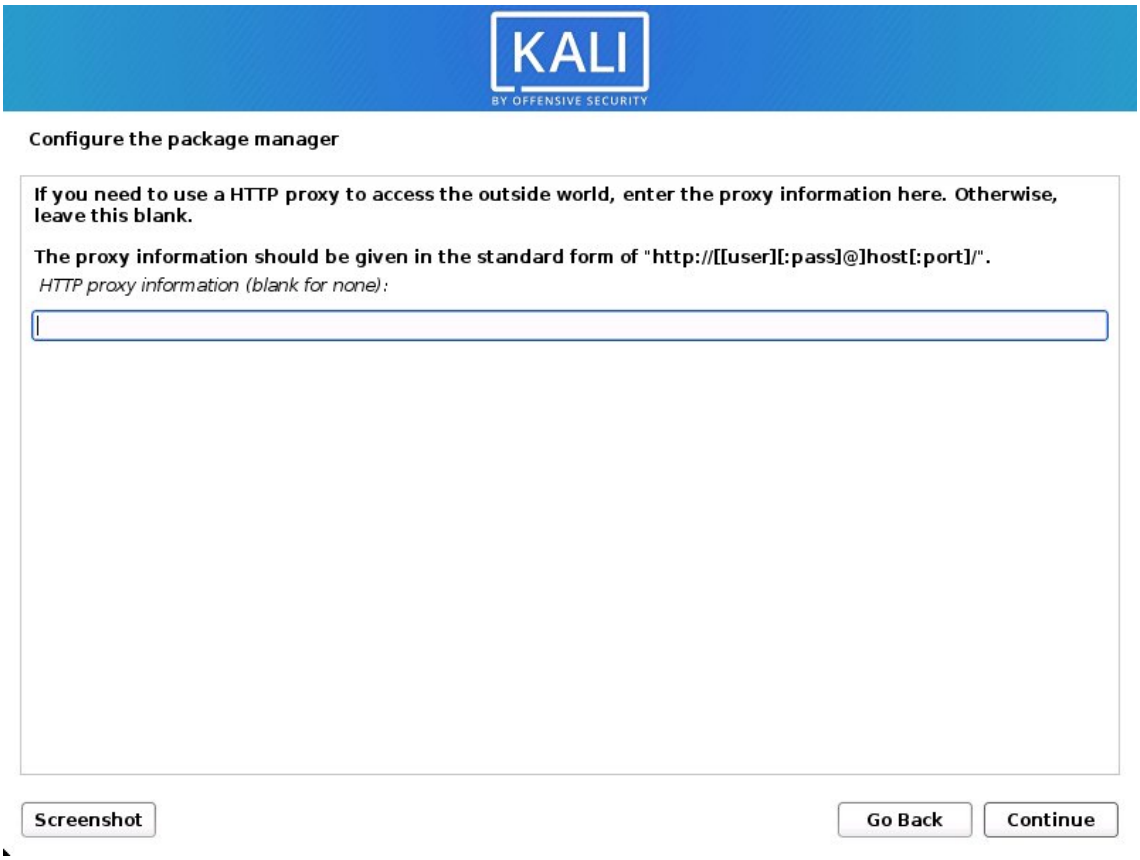

Figure 6.15 — HTTP proxy: leave blank.

6.10 Software Selection

Step 38: Keep the default selections: Xfce desktop environment, top10 tools, and default tools. Click Continue.

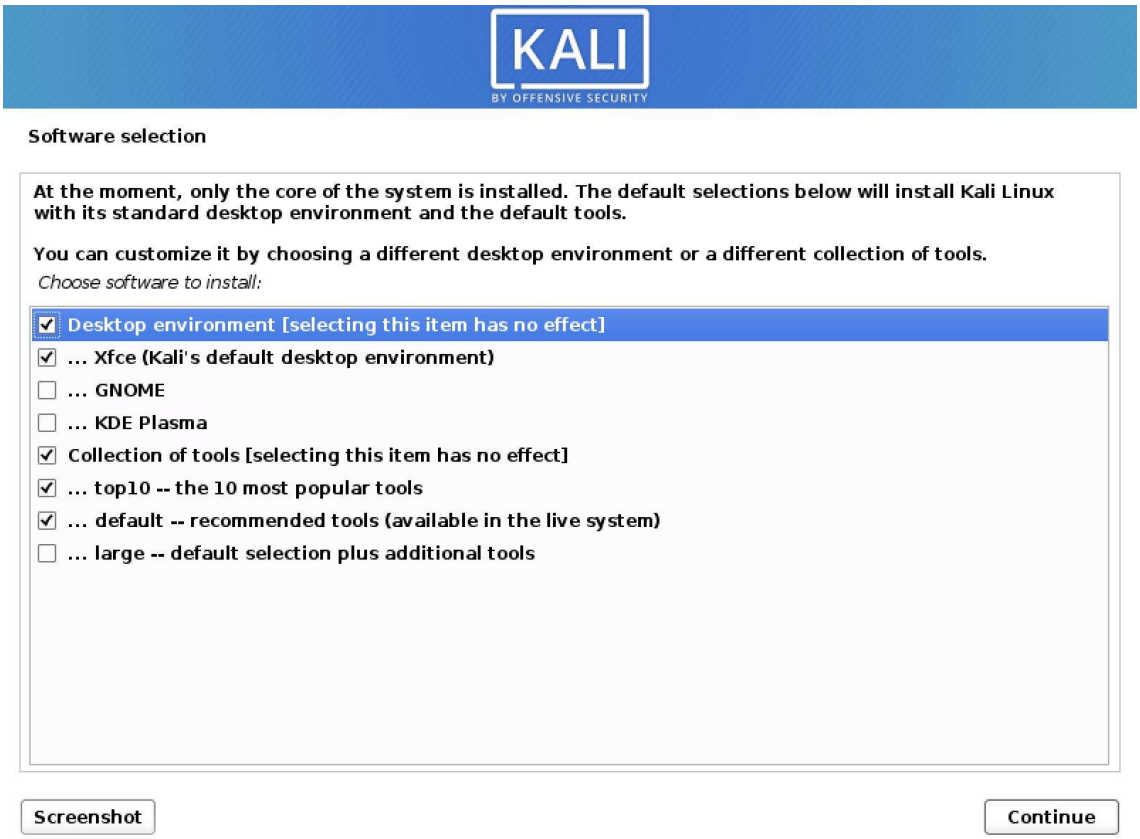

Figure 6.16 — Software selection with defaults.

Understanding the Options

Xfce: Lightweight, fast desktop — perfect for VMs. | top10: The 10 most popular tools (Nmap, Metasploit, Burp Suite, etc). | default: Broader recommended tool set. | large: Everything + specialized tools (uses more disk space and time).

Installation now begins copying files and installing packages. This takes 10-20 minutes. A progress bar shows the current status.

6.11 GRUB Boot Loader

Step 39: Select "Yes" to install GRUB to the primary drive. Click Continue.

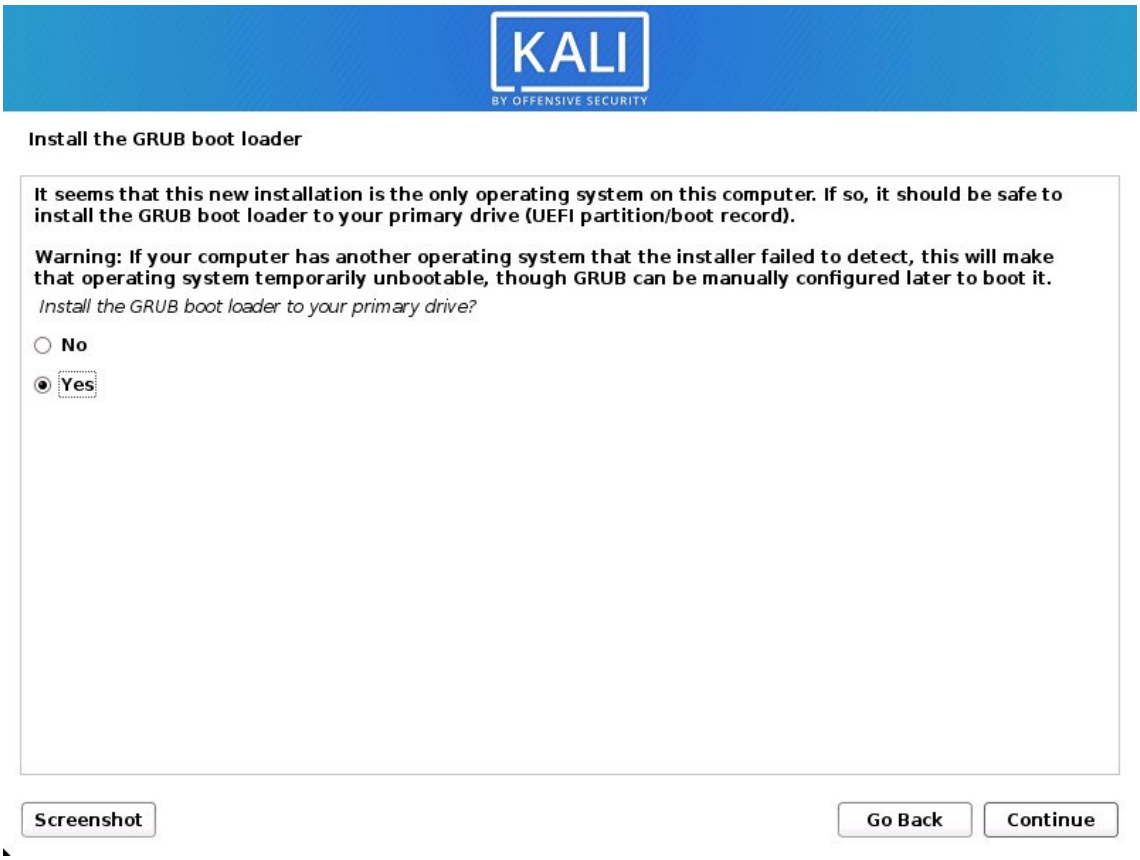

Figure 6.17 — Install GRUB: select Yes.

> **Fun Fact!**
> GRUB (GRand Unified Bootloader) is the first software that runs when Linux starts. It loads the kernel into memory and hands over control. Without GRUB, your system literally does not know how to start the operating system.

Step 40: Select /dev/sda as the GRUB installation device. Do NOT select "Enter device manually."

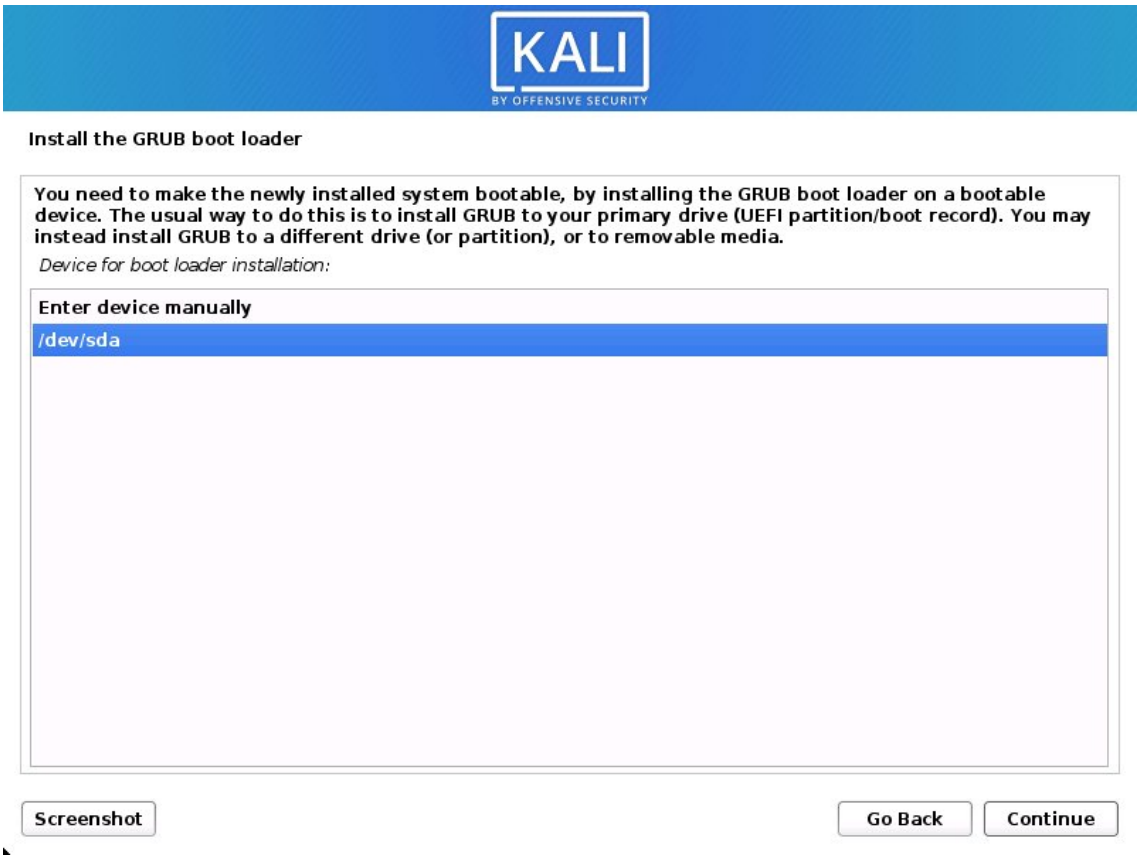

Figure 6.18 — Select /dev/sda for GRUB.

Always Select /dev/sda

Skipping this or selecting the wrong device means the VM cannot boot after installation. Always pick the /dev/sda option from the list.

6.12 Installation Complete!

Step 41: Click "Continue" to reboot into your freshly installed Kali Linux!

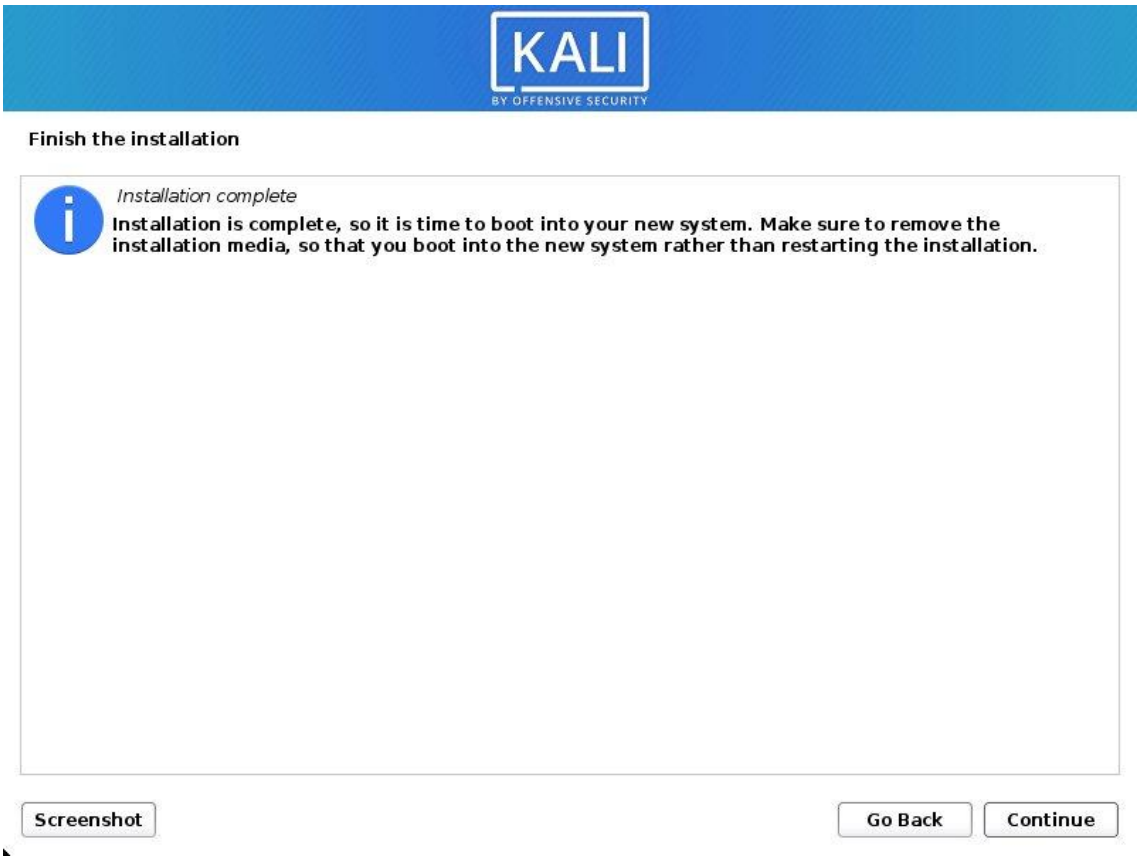

Figure 6.19 — Installation complete! Reboot to start using Kali.

After rebooting, the GRUB boot menu appears briefly (auto-selects Kali), then the login screen loads. Enter your username and password to log in.

> **First Login**
> After logging in, you will see the Kali Xfce desktop. Open a terminal by right-clicking the desktop > "Open Terminal Here" or from the top menu bar. This is your command center for all security tools.

7. Post-Installation Setup

After first login, perform these essential steps to ensure your Kali system is current and fully functional.

7.1 Update the System

Open a terminal and run:

```
$ sudo apt update && sudo apt full-upgrade -y
```

"apt update" refreshes the package list from Kali repositories. "apt full-upgrade" downloads and installs all available updates. "-y" auto-confirms.

7.2 Verify Kali Version

```
$ cat /etc/os-release
```

This shows your Kali version, codename, and build information.

7.3 Install Additional Tools (Optional)

```
$ sudo apt install -y kali-linux-large
```

Installs the "large" metapackage with hundreds of additional security tools. Requires several GB of disk space and download time.

7.4 Take a Snapshot

With Kali freshly installed and updated, take a VirtualBox snapshot. Go to Machine > Take Snapshot (Ctrl+T). Name it "Fresh Install - Updated." This lets you restore to this clean state anytime.

> **Why Snapshots Matter**
> During security labs, you may install vulnerable software, change configs, or break something. Snapshots let you instantly restore your VM to a known-good state. Think of it as a save point in a video game — essential for any serious lab work.

8. Common Issues & Troubleshooting

8.1 VT-x/AMD-V Not Available

Error: "VT-x is not available (VERR_VMX_NO_VMX)"

Solution: Restart > Enter BIOS (F2/F10/DEL/ESC) > Enable Intel VT-x or AMD-V > Save & Exit.

8.2 VM is Very Slow

Fixes: Increase RAM to 4096 MB (Settings > System). Increase CPUs to 2 (Settings > System > Processor). Set Video Memory to 128 MB (Settings > Display). Disable 3D Acceleration. Close heavy host applications.

8.3 Black Screen After Boot

Disable 3D Acceleration: Settings > Display > uncheck "Enable 3D Acceleration." Some graphics drivers conflict with VirtualBox rendering.

8.4 No Internet Inside Kali

Check: Settings > Network > Adapter 1 = "NAT" (default). Inside Kali terminal: sudo dhclient. Test: ping google.com

8.5 VERR_PATH_NOT_FOUND Error

The virtual disk file (.vdi) may have been moved or is on OneDrive/synced folder. Move your VirtualBox VMs folder to a local non-synced directory (e.g., C:\VirtualBox VMs\).

8.6 ISO Opens with WinRAR

You do not need to open or extract the ISO. The ISO is used by VirtualBox directly through VM settings (Section 4.3). Never extract an ISO for VirtualBox use.

8.7 Shared Clipboard Not Working

Install VirtualBox Guest Additions inside Kali: Devices menu > Insert Guest Additions CD Image > open terminal > run: sudo apt install -y virtualbox-guest-x11. Then reboot the VM.

9. Quick Reference Card

Item	Details
Kali Download	kali.org/get-kali
VirtualBox Download	virtualbox.org/wiki/Downloads
Default VM Name	Kali-Linux
Recommended RAM	4096 MB (4 GB)
Recommended CPUs	2
Recommended Disk	80 GB (dynamically allocated)
Video Memory	128 MB
OS Type in VirtualBox	Linux / Debian or Ubuntu (64-bit)
Default Credentials (Pre-built)	kali / kali
Update Command	sudo apt update && sudo apt full-upgrade -y
VirtualBox Host Key	Right Ctrl (releases mouse)
Take Snapshot	Machine > Take Snapshot (Ctrl+T)
Guest Additions	sudo apt install virtualbox-guest-x11

10. Conclusion

Congratulations! You have successfully installed Kali Linux on Oracle VirtualBox. You now have a fully functional penetration testing environment ready for your NIAI cybersecurity labs.

In this guide you learned how to download Kali Linux and VirtualBox from official sources, create and configure a virtual machine with appropriate hardware settings, attach an ISO and boot from it, navigate the complete Kali installation process (language, network, user accounts, disk partitioning, software selection, bootloader), and perform post-installation updates and snapshots.

Remember to take regular snapshots before experimenting, keep your system updated, and never use Kali tools against systems without explicit written permission.

Ethical Use Disclaimer

Kali Linux contains powerful security testing tools. Using these tools against systems without explicit written authorization is illegal and unethical. Always obtain proper permission before conducting any security assessment. This lab environment is for educational purposes within the NIAI program only.

Developed under the NIAI Program
NAVTTC × NETSOL Technologies — 2026